CALIFORNIA REGIONAL WATER QUALITY CONTROL BOARD

CENTRAL VALLEY REGION
11020 SUN CENTER DR #200
RANCHO CORDOVA CA 95670-6114

2023 ANNUAL REPORT

PREPARED FOR:

SOARES DAIRY #1

11706 VINCENT RD

TURLOCK CA 95380

NOTES:

PREPARED BY:

CAS
CARDOSO AG SERVICES

PO BOX 906
NEWMAN CA 95360

Annual Report - General Order No. R5-2007-0035
Reporting period 01/01/2023 to 12/31/2023.

DAIRY FACILITY INFORMATION

A. NAME OF DAIRY OR BUSINESS OPERATING THE DAIRY: Joe & Teresa Soares Dairy #1

Physical address of dairy:

Number and Street	City	County	Zip Code
11076 Vincent RD	Turlock	Merced	95380

Street and nearest cross street (if no address):

Date facility was originally placed in operation: 01/01/1950

Regional Water Quality Control Board Basin Plan designation: San Joaquin River Basin

County Assessor Parcel Number(s) for dairy facility:

X041-X162-X003-XXXX

B. OPERATORS

Soares, Jose

Operator name: Soares, Jose

Telephone no.: (209) 652-2990 (Landline) (Cellular)

Mailing Address Number and Street	City	State	Zip Code
P.O. Box 189	Delhi	CA	95135

This operator is responsible for paying permit fees.

C. OWNERS

Soares, Jose

Legal owner name: Soares, Jose

Telephone no.: (209) 652-2990 (Landline) (Cellular)

Mailing Address Number and Street	City	State	Zip Code
P.O. Box 189	Delhi	CA	95135

This owner is responsible for paying permit fees.

AVAILABLE NUTRIENTS

A. HERD INFORMATION

	Milk Cows	Dry Cows	Bred Heifers (15-24 mo.)	Heifers (7-14 mo. to breeding)	Calves (4-6 mo.)	Calves (0-3 mo.)
Number open confinement	0	0	0	225	670	1,600
Number under roof	1,125	230	0	0	0	0
Maximum number	1,125	230	0	225	670	1,600
Average number	1,125	230	0	225	670	1,600
Avg live weight (lbs)	1,400	1,500	0	775		

Predominant milk cow breed: Holstein

Average milk production: 76 *pounds per cow per day*

B. MANURE GENERATED

Total manure excreted by the herd: 43,118.53 *tons per reporting period*

Total nitrogen from manure: 566,984.72 *lbs per reporting period*

After ammonia losses (30% loss applied): 396,889.30 *lbs per reporting period*

Total phosphorus from manure: 83,136.39 *lbs per reporting period*

Total potassium from manure: 219,882.63 *lbs per reporting period*

Total salt from manure: 582,594.75 *lbs per reporting period*

C. PROCESS WASTEWATER GENERATED

Process wastewater generated: 12,825,000 *gallons*

Total nitrogen generated: 67,883.52 *lbs*

Total phosphorus generated: 8,555.41 *lbs*

Total potassium generated: 88,172.33 *lbs*

Total salt generated: 485,689.01 *lbs*

	12,825,000	*gallons applied*
+	0	*gallons exported*
-	0	*gallons imported*
=	12,825,000	*gallons generated*

D. FRESH WATER SOURCES

Source Description	Type
Canal	Surface water
Domestic Well	Ground water

E. SUBSURFACE (TILE) DRAINAGE SOURCES

No subsurface (tile) drainage sources entered.

F. NUTRIENT IMPORTS

No dry manure nutrient imports entered.

No process wastewater nutrient imports entered.

No commercial or other nutrient imports entered.

G. NUTRIENT EXPORTS

Date	Material type	Quantity	Reporting basis	Moisture (%)	Density (lbs/cu ft)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
05/08/2023	Separator solids	4,095.00 *ton*	Dry-weight	75.3		27,600.00	7,600.00	17,200.00		0.00
11/14/2023	Separator solids	4,160.00 *ton*	Dry-weight	72.9		31,900.00	7,200.00	14,100.00		0.00

No liquid nutrient exports entered.

Material type	Total N (lbs)	Total P (lbs)	Total K (lbs)	Total salt (lbs)
Dry manure	127,758.44	31,608.25	66,585.95	0.00
Process wastewater	0.00	0.00	0.00	0.00
Total exports for all materials	127,758.44	31,608.25	66,585.95	0.00

APPLICATION AREA

A. LIST OF LAND APPLICATION AREAS

Field name	Controlled acres	Cropable acres	Total harvests	Type of waste applied	Parcel number
1	9	9	2	process wastewater	X041-X162-X003-XXXX
2	21	21	2	process wastewater	X041-X162-X003-XXXX
3	22	22	2	process wastewater	X041-X162-X003-XXXX
4	43	43	2	process wastewater	X041-X162-X003-XXXX
5	12	12	2	process wastewater	X041-X162-X003-XXXX
C1	31	31	2	manure	X046-X030-X008-XXXX
C2	41	41	2	manure	X046-X030-X008-XXXX
H1	20	20	2	manure	X046-X030-X007-XXXX
M1	30	30	2	process wastewater	X046-X030-X006-XXXX
M2	28	28	2	process wastewater	X046-X030-X006-XXXX
N1	16	16	1	none	X046-X040-X025-XXXX
Totals for areas that were used for application	257	257	20		
Totals for areas that were not used for application	16	16	1		
Land application area totals	273	273	21		

B. CROPS AND HARVESTS

1

Field name: 1

09/10/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 9 Plant date: 09/10/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
05/10/2023	136.00 *ton*	Dry-weight		78.4	23,300.00	3,200.00	27,100.00		8.30

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	16.00	160.00	25.60	132.80	0.00
Total actual harvest content	15.11	152.10	20.89	176.91	541.82

1

05/22/2023: Corn, silage

Crop: Corn, silage Acres planted: 9 Plant date: 05/22/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
09/28/2023	214.00 *ton*	Dry-weight		66.6	14,900.00	3,400.00	12,600.00		6.10

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	30.00	240.00	45.00	198.00	0.00
Total actual harvest content	23.78	236.66	54.00	200.13	968.90

2

Field name: 2

09/10/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 21 Plant date: 09/10/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
05/10/2023	317.00 *ton*	Dry-weight		78.0	24,400.00	4,200.00	32,000.00		9.80

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	16.00	160.00	25.60	132.80	0.00
Total actual harvest content	15.10	162.06	27.90	212.54	650.91

05/22/2023: Corn, silage

Crop: Corn, silage Acres planted: 21 Plant date: 05/22/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
10/02/2023	499.00 *ton*	Dry-weight		67.7	13,100.00	2,800.00	12,800.00		4.70

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	30.00	240.00	45.00	198.00	0.00
Total actual harvest content	23.76	201.09	42.98	196.48	721.46

3

Field name: 3

09/10/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 22 Plant date: 09/10/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
05/10/2023	332.00 *ton*	Dry-weight		76.6	20,900.00	4,300.00	29,300.00		9.20

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	16.00	160.00	25.60	132.80	0.00
Total actual harvest content	15.09	147.61	30.37	206.93	649.75

05/22/2023: Corn, silage

Crop: Corn, silage Acres planted: 22 Plant date: 05/22/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
10/02/2023	523.00 *ton*	Dry-weight		67.4	14,300.00	3,300.00	13,200.00		5.60

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	30.00	240.00	45.00	198.00	0.00
Total actual harvest content	23.77	221.65	51.15	204.60	867.99

4

Field name: 4

09/10/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 43 Plant date: 09/10/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
05/10/2023	649.00 *ton*	Dry-weight		70.5	15,500.00	3,200.00	27,000.00		7.30

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	16.00	160.00	25.60	132.80	0.00
Total actual harvest content	15.09	138.03	28.50	240.43	650.06

4

05/22/2023: Corn, silage

Crop: Corn, silage Acres planted: 43 Plant date: 05/22/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
09/29/2023	1,022.00 *ton*	Dry-weight		69.4	15,900.00	3,200.00	17,900.00		6.10

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	30.00	240.00	45.00	198.00	0.00
Total actual harvest content	23.77	231.28	46.55	260.37	887.29

5

Field name: 5

09/10/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 12 Plant date: 09/10/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
05/09/2023	181.00 *ton*	Dry-weight		69.8	21,400.00	4,000.00	33,200.00		10.40

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	16.00	160.00	25.60	132.80	0.00
Total actual harvest content	15.08	194.96	36.44	302.46	947.47

05/22/2023: Corn, silage

Crop: Corn, silage Acres planted: 12 Plant date: 05/22/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
09/28/2023	285.00 *ton*	Dry-weight		61.1	11,300.00	3,100.00	13,000.00		5.70

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	30.00	240.00	45.00	198.00	0.00
Total actual harvest content	23.75	208.80	57.28	240.21	1,053.22

C1

Field name: C1

09/15/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 31 Plant date: 09/15/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
05/08/2023	431.00 *ton*	Dry-weight		66.1	17,200.00	3,100.00	21,400.00		10.75

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	16.00	160.00	25.60	132.80	0.00
Total actual harvest content	13.90	162.13	29.22	201.72	1,013.34

05/26/2023: Corn, silage

Crop: Corn, silage Acres planted: 31 Plant date: 05/26/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
09/29/2023	643.00 *ton*	Dry-weight		65.1	14,900.00	2,800.00	14,400.00		5.20

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	30.00	240.00	45.00	198.00	0.00
Total actual harvest content	20.74	215.72	40.54	208.48	752.85

C2

Field name: C2

09/15/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 41 Plant date: 09/15/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
05/09/2023	570.00 *ton*	Dry-weight		69.6	21,750.00	3,450.00	28,550.00		10.20

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	16.00	160.00	25.60	132.80	0.00
Total actual harvest content	13.90	183.85	29.16	241.32	862.17

C2

05/26/2023: Corn, silage

Crop: Corn, silage Acres planted: 41 Plant date: 05/26/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
09/29/2023	850.00 *ton*	Dry-weight		64.5	15,100.00	3,300.00	21,100.00		6.80

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	30.00	240.00	45.00	198.00	0.00
Total actual harvest content	20.73	222.26	48.57	310.58	1,000.93

H1

Field name: H1

09/11/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 20 Plant date: 09/11/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
05/08/2023	274.00 *ton*	Dry-weight		72.7	25,100.00	4,200.00	33,800.00		10.00

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	16.00	160.00	25.60	132.80	0.00
Total actual harvest content	13.70	187.75	31.42	252.83	748.02

05/26/2023: Corn, silage

Crop: Corn, silage Acres planted: 20 Plant date: 05/26/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
09/29/2023	449.00 *ton*	Dry-weight		73.6	16,200.00	2,900.00	21,300.00		6.90

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	30.00	240.00	45.00	198.00	0.00
Total actual harvest content	22.45	192.03	34.38	252.48	817.90

M1

Field name: M1

09/12/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 30 Plant date: 09/12/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
05/08/2023	426.00 *ton*	Dry-weight		70.9	22,650.00	3,950.00	31,950.00		13.60

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	16.00	160.00	25.60	132.80	0.00
Total actual harvest content	14.20	187.19	32.64	264.05	1,123.96

05/25/2023: Corn, silage

Crop: Corn, silage Acres planted: 30 Plant date: 05/25/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
09/29/2023	716.00 *ton*	Dry-weight		66.0	15,200.00	3,000.00	19,800.00		6.40

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	30.00	240.00	45.00	198.00	0.00
Total actual harvest content	23.87	246.69	48.69	321.34	1,038.68

M2

Field name: M2

09/12/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 28 Plant date: 09/12/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
05/08/2023	398.00 *ton*	Dry-weight		70.9	23,500.00	3,850.00	34,250.00		10.70

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	16.00	160.00	25.60	132.80	0.00
Total actual harvest content	14.21	194.41	31.85	283.34	885.18

M2

05/25/2023: Corn, silage

Crop: Corn, silage Acres planted: 28 Plant date: 05/25/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
09/29/2023	668.00 *ton*	Dry-weight		69.2	17,500.00	3,000.00	14,900.00		5.90

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	30.00	240.00	45.00	198.00	0.00
Total actual harvest content	23.86	257.18	44.09	218.97	867.06

N1

Field name: N1

01/12/2016: Almond, in shell

Crop: Almond, in shell Acres planted: 16 Plant date: 01/12/2016

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
10/08/2023	11.00 *ton*	Dry-weight		6.1	24,400.00	2,600.00	13,800.00		1.19

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	1.50	195.00	30.00	150.00	0.00
Total actual harvest content	0.69	31.50	3.36	17.82	15.36

NUTRIENT BUDGET

A. LAND APPLICATIONS

1 - 09/10/2022: Oats, silage-soft dough

Field name: 1

Crop: Oats, silage-soft dough Plant date: 09/10/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/06/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	84.70	8.54	141.87	639.78	150,000.00 *gal*
Canal	Surface water	15.47	0.00	0.00	0.00	2,606,437.00 *gal*
Application event totals		100.17	8.54	141.87	639.78	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/12/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	95.87	9.70	106.12	564.68	105,000.00 *gal*
Canal	Surface water	4.21	0.00	0.00	0.00	709,407.00 *gal*
Application event totals		100.08	9.70	106.12	564.68	

1 - 05/22/2023: Corn, silage

Field name: 1

Crop: Corn, silage Plant date: 05/22/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/20/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	6.69	0.00	0.00	0.00	1,126,708.00 *gal*
Application event totals		6.69	0.00	0.00	0.00	

1 - 05/22/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/24/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	6.69	0.00	0.00	0.00	1,126,708.00 *gal*
Application event totals		6.69	0.00	0.00	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/07/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	74.49	10.93	85.95	560.51	150,000.00 *gal*
Canal	Surface water	8.42	0.00	0.00	0.00	1,418,817.00 *gal*
Application event totals		82.91	10.93	85.95	560.51	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/20/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	6.77	0.00	0.00	0.00	1,140,894.00 *gal*
Application event totals		6.77	0.00	0.00	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/30/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	67.04	9.84	77.36	504.46	135,000.00 *gal*
Canal	Surface water	6.19	0.00	0.00	0.00	1,043,249.00 *gal*
Application event totals		73.23	9.84	77.36	504.46	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/11/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	59.59	8.75	68.76	448.40	120,000.00 *gal*
Canal	Surface water	6.44	0.00	0.00	0.00	1,084,978.00 *gal*
Application event totals		66.03	8.75	68.76	448.40	

1 - 05/22/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/21/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	9.41	0.00	0.00	0.00	1,585,738.00 *gal*
Application event totals		9.41	0.00	0.00	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/03/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	11.39	0.00	0.00	0.00	1,919,574.00 *gal*
Application event totals		11.39	0.00	0.00	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/15/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	11.39	0.00	0.00	0.00	1,919,574.00 *gal*
Application event totals		11.39	0.00	0.00	0.00	

2 - 09/10/2022: Oats, silage-soft dough

Field name: 2

Crop: Oats, silage-soft dough

Plant date: 09/10/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/06/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	90.75	9.15	152.00	685.48	375,000.00 *gal*
Canal	Surface water	15.47	0.00	0.00	0.00	6,081,687.00 *gal*
Application event totals		106.22	9.15	152.00	685.48	

2 - 09/10/2022: Oats, silage-soft dough

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/12/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	93.91	9.50	103.95	553.15	240,000.00 *gal*
Canal	Surface water	4.21	0.00	0.00	0.00	1,655,284.00 *gal*
Application event totals		98.12	9.50	103.95	553.15	

2 - 05/22/2023: Corn, silage

Field name: 2

Crop: Corn, silage

Plant date: 05/22/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/20/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	6.69	0.00	0.00	0.00	2,628,985.00 *gal*
Application event totals		6.69	0.00	0.00	0.00	

06/24/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	6.69	0.00	0.00	0.00	2,628,985.00 *gal*
Application event totals		6.69	0.00	0.00	0.00	

07/07/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	73.43	10.78	84.73	552.50	345,000.00 *gal*
Canal	Surface water	8.42	0.00	0.00	0.00	3,310,574.00 *gal*
Application event totals		81.85	10.78	84.73	552.50	

2 - 05/22/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/20/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	6.77	0.00	0.00	0.00	2,662,085.00 *gal*
Application event totals		6.77	0.00	0.00	0.00	

07/30/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	67.04	9.84	77.36	504.46	315,000.00 *gal*
Canal	Surface water	6.19	0.00	0.00	0.00	2,434,247.00 *gal*
Application event totals		73.23	9.84	77.36	504.46	

08/11/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	60.66	8.90	69.99	456.41	285,000.00 *gal*
Canal	Surface water	6.44	0.00	0.00	0.00	2,531,616.00 *gal*
Application event totals		67.10	8.90	69.99	456.41	

08/21/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	9.41	0.00	0.00	0.00	3,700,055.00 *gal*
Application event totals		9.41	0.00	0.00	0.00	

09/03/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	11.39	0.00	0.00	0.00	4,479,006.00 *gal*
Application event totals		11.39	0.00	0.00	0.00	

09/15/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	11.39	0.00	0.00	0.00	4,479,006.00 *gal*
Application event totals		11.39	0.00	0.00	0.00	

3 - 09/10/2022: Oats, silage-soft dough

Field name: 3

Crop: Oats, silage-soft dough

Plant date: 09/10/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/06/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	76.23	7.69	127.68	575.81	330,000.00 *gal*
Canal	Surface water	15.47	0.00	0.00	0.00	6,371,291.00 *gal*
Application event totals		91.70	7.69	127.68	575.81	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/12/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	95.25	9.63	105.43	561.01	255,000.00 *gal*
Canal	Surface water	4.21	0.00	0.00	0.00	1,734,107.00 *gal*
Application event totals		99.46	9.63	105.43	561.01	

3 - 05/22/2023: Corn, silage

Field name: 3

Crop: Corn, silage

Plant date: 05/22/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/20/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	6.69	0.00	0.00	0.00	2,754,175.00 *gal*
Application event totals		6.69	0.00	0.00	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/24/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	6.69	0.00	0.00	0.00	2,754,175.00 *gal*
Application event totals		6.69	0.00	0.00	0.00	

3 - 05/22/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/07/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	85.33	12.52	98.46	642.03	420,000.00 *gal*
Canal	Surface water	8.42	0.00	0.00	0.00	3,468,220.00 *gal*
Application event totals		93.75	12.52	98.46	642.03	

07/20/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	6.77	0.00	0.00	0.00	2,788,851.00 *gal*
Application event totals		6.77	0.00	0.00	0.00	

07/30/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	73.14	10.73	84.39	550.31	360,000.00 *gal*
Canal	Surface water	6.19	0.00	0.00	0.00	2,550,164.00 *gal*
Application event totals		79.33	10.73	84.39	550.31	

08/11/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	67.04	9.84	77.36	504.46	330,000.00 *gal*
Canal	Surface water	6.44	0.00	0.00	0.00	2,652,169.00 *gal*
Application event totals		73.48	9.84	77.36	504.46	

08/21/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	9.41	0.00	0.00	0.00	3,876,249.00 *gal*
Application event totals		9.41	0.00	0.00	0.00	

3 - 05/22/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/03/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	11.39	0.00	0.00	0.00	4,692,292.00 *gal*
Application event totals		11.39	0.00	0.00	0.00	

09/15/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	11.39	0.00	0.00	0.00	4,692,292.00 *gal*
Application event totals		11.39	0.00	0.00	0.00	

4 - 09/10/2022: Oats, silage-soft dough

Field name: 4

Crop: Oats, silage-soft dough Plant date: 09/10/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/06/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	70.91	7.15	118.77	535.63	600,000.00 *gal*
Canal	Surface water	15.47	0.00	0.00	0.00	12,452,977.00 *gal*
Application event totals		86.38	7.15	118.77	535.63	

04/12/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	94.59	9.57	104.71	557.17	495,000.00 *gal*
Canal	Surface water	4.21	0.00	0.00	0.00	3,389,390.00 *gal*
Application event totals		98.80	9.57	104.71	557.17	

4 - 05/22/2023: Corn, silage

4 - 05/22/2023: Corn, silage

Field name: 4

Crop: Corn, silage

Plant date: 05/22/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/20/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	6.69	0.00	0.00	0.00	5,383,160.00 *gal*
Application event totals		6.69	0.00	0.00	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/24/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	6.69	0.00	0.00	0.00	5,383,160.00 *gal*
Application event totals		6.69	0.00	0.00	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/07/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	87.31	12.81	100.75	656.96	840,000.00 *gal*
Canal	Surface water	8.42	0.00	0.00	0.00	6,778,793.00 *gal*
Application event totals		95.73	12.81	100.75	656.96	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/20/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	6.77	0.00	0.00	0.00	5,450,936.00 *gal*
Application event totals		6.77	0.00	0.00	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/30/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	74.84	10.98	86.35	563.11	720,000.00 *gal*
Canal	Surface water	6.19	0.00	0.00	0.00	4,984,411.00 *gal*
Application event totals		81.03	10.98	86.35	563.11	

4 - 05/22/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/11/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	68.60	10.07	79.16	516.19	660,000.00 *gal*
Canal	Surface water	6.44	0.00	0.00	0.00	5,183,786.00 *gal*
Application event totals		75.04	10.07	79.16	516.19	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/21/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	9.41	0.00	0.00	0.00	7,576,304.00 *gal*
Application event totals		9.41	0.00	0.00	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/03/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	11.39	0.00	0.00	0.00	9,171,299.00 *gal*
Application event totals		11.39	0.00	0.00	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/15/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	11.39	0.00	0.00	0.00	9,171,299.00 *gal*
Application event totals		11.39	0.00	0.00	0.00	

5 - 09/10/2022: Oats, silage-soft dough

Field name: 5

Crop: Oats, silage-soft dough

Plant date: 09/10/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following

5 - 09/10/2022: Oats, silage-soft dough

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/06/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	95.29	9.61	159.60	719.76	225,000.00 *gal*
Canal	Surface water	15.47	0.00	0.00	0.00	3,475,249.00 *gal*
Application event totals		110.76	9.61	159.60	719.76	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/12/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	102.72	10.39	113.70	605.01	150,000.00 *gal*
Canal	Surface water	4.21	0.00	0.00	0.00	945,876.00 *gal*
Application event totals		106.93	10.39	113.70	605.01	

5 - 05/22/2023: Corn, silage

Field name: 5

Crop: Corn, silage

Plant date: 05/22/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/20/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	6.69	0.00	0.00	0.00	1,502,277.00 *gal*
Application event totals		6.69	0.00	0.00	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/24/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	6.69	0.00	0.00	0.00	1,502,277.00 *gal*
Application event totals		6.69	0.00	0.00	0.00	

5 - 05/22/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/07/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	72.63	10.66	83.80	546.49	195,000.00 *gal*
Canal	Surface water	8.42	0.00	0.00	0.00	1,891,756.00 *gal*
Application event totals		81.05	10.66	83.80	546.49	

07/20/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	6.77	0.00	0.00	0.00	1,521,192.00 *gal*
Application event totals		6.77	0.00	0.00	0.00	

07/30/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	67.04	9.84	77.36	504.46	180,000.00 *gal*
Canal	Surface water	6.19	0.00	0.00	0.00	1,390,998.00 *gal*
Application event totals		73.23	9.84	77.36	504.46	

08/11/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	61.46	9.02	70.91	462.42	165,000.00 *gal*
Canal	Surface water	6.44	0.00	0.00	0.00	1,446,638.00 *gal*
Application event totals		67.90	9.02	70.91	462.42	

08/21/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	9.41	0.00	0.00	0.00	2,114,317.00 *gal*
Application event totals		9.41	0.00	0.00	0.00	

5 - 05/22/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/03/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	11.39	0.00	0.00	0.00	2,559,432.00 *gal*
Application event totals		11.39	0.00	0.00	0.00	

09/15/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	11.39	0.00	0.00	0.00	2,559,432.00 *gal*
Application event totals		11.39	0.00	0.00	0.00	

C1 - 09/15/2022: Oats, silage-soft dough

Field name: C1

Crop: Oats, silage-soft dough

Plant date: 09/15/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/01/2022	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Solid Manure	Separator solids	199.67	45.07	88.26	0.00	358.00 *ton*
Application event totals		199.67	45.07	88.26	0.00	

10/06/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	15.47	0.00	0.00	0.00	8,977,728.00 *gal*
Application event totals		15.47	0.00	0.00	0.00	

04/11/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	5.09	0.00	0.00	0.00	2,957,119.00 *gal*
Application event totals		5.09	0.00	0.00	0.00	

C1 - 05/26/2023: Corn, silage

Field name: C1

Crop: Corn, silage

Plant date: 05/26/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/10/2023	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Solid Manure	Separator solids	214.63	59.10	133.76	0.00	488.00 *ton*
Application event totals		214.63	59.10	133.76	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/17/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	8.15	0.00	0.00	0.00	4,731,385.00 *gal*
Application event totals		8.15	0.00	0.00	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/22/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	8.15	0.00	0.00	0.00	4,731,385.00 *gal*
Application event totals		8.15	0.00	0.00	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/05/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	9.51	0.00	0.00	0.00	5,519,944.00 *gal*
Application event totals		9.51	0.00	0.00	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/17/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	10.90	0.00	0.00	0.00	6,324,289.00 *gal*
Application event totals		10.90	0.00	0.00	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/28/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	8.49	0.00	0.00	0.00	4,928,531.00 *gal*
Application event totals		8.49	0.00	0.00	0.00	

C1 - 05/26/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/09/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	8.83	0.00	0.00	0.00	5,125,664.00 *gal*
Application event totals		8.83	0.00	0.00	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/20/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	8.83	0.00	0.00	0.00	5,125,664.00 *gal*
Application event totals		8.83	0.00	0.00	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/01/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	8.83	0.00	0.00	0.00	5,125,664.00 *gal*
Application event totals		8.83	0.00	0.00	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/13/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	8.83	0.00	0.00	0.00	5,125,664.00 *gal*
Application event totals		8.83	0.00	0.00	0.00	

C2 - 09/15/2022: Oats, silage-soft dough

Field name: C2

Crop: Oats, silage-soft dough

Plant date: 09/15/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/01/2022	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Solid Manure	Separator solids	229.83	51.87	101.59	0.00	545.00 *ton*
Application event totals		229.83	51.87	101.59	0.00	

C2 - 09/15/2022: Oats, silage-soft dough

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/06/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	15.47	0.00	0.00	0.00	11,873,769.00 *gal*
Application event totals		15.47	0.00	0.00	0.00	

04/11/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	5.09	0.00	0.00	0.00	3,911,028.00 *gal*
Application event totals		5.09	0.00	0.00	0.00	

C2 - 05/26/2023: Corn, silage

Field name: C2

Crop: Corn, silage

Plant date: 05/26/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/10/2023	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Solid Manure	Separator solids	223.80	61.63	139.47	0.00	673.00 *ton*
Application event totals		223.80	61.63	139.47	0.00	

05/17/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	8.15	0.00	0.00	0.00	6,257,638.00 *gal*
Application event totals		8.15	0.00	0.00	0.00	

06/22/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	8.15	0.00	0.00	0.00	6,257,638.00 *gal*
Application event totals		8.15	0.00	0.00	0.00	

C2 - 05/26/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/05/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	9.51	0.00	0.00	0.00	7,300,571.00 *gal*
Application event totals		9.51	0.00	0.00	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/17/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	10.90	0.00	0.00	0.00	8,364,382.00 *gal*
Application event totals		10.90	0.00	0.00	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/28/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	8.49	0.00	0.00	0.00	6,518,380.00 *gal*
Application event totals		8.49	0.00	0.00	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/09/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	8.83	0.00	0.00	0.00	6,779,104.00 *gal*
Application event totals		8.83	0.00	0.00	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/20/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	8.83	0.00	0.00	0.00	6,779,104.00 *gal*
Application event totals		8.83	0.00	0.00	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/01/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	8.83	0.00	0.00	0.00	6,779,104.00 *gal*
Application event totals		8.83	0.00	0.00	0.00	

C2 - 05/26/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/13/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	8.83	0.00	0.00	0.00	6,779,104.00 *gal*
Application event totals		8.83	0.00	0.00	0.00	

H1 - 09/11/2022: Oats, silage-soft dough

Field name: H1

Crop: Oats, silage-soft dough

Plant date: 09/11/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/01/2022	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Solid Manure	Separator solids	233.41	52.68	103.17	0.00	270.00 *ton*
Application event totals		233.41	52.68	103.17	0.00	

10/06/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	15.47	0.00	0.00	0.00	5,792,082.00 *gal*
Application event totals		15.47	0.00	0.00	0.00	

04/11/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	5.74	0.00	0.00	0.00	2,150,679.00 *gal*
Application event totals		5.74	0.00	0.00	0.00	

H1 - 05/26/2023: Corn, silage

Field name: H1

Crop: Corn, silage

Plant date: 05/26/2023

H1 - 05/26/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/10/2023	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Solid Manure	Separator solids	182.02	50.12	113.43	0.00	267.00 *ton*
Application event totals		182.02	50.12	113.43	0.00	

05/19/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	6.89	0.00	0.00	0.00	2,580,802.00 *gal*
Application event totals		6.89	0.00	0.00	0.00	

06/23/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	6.89	0.00	0.00	0.00	2,580,802.00 *gal*
Application event totals		6.89	0.00	0.00	0.00	

07/06/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	8.04	0.00	0.00	0.00	3,010,941.00 *gal*
Application event totals		8.04	0.00	0.00	0.00	

07/20/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	8.80	0.00	0.00	0.00	3,296,984.00 *gal*
Application event totals		8.80	0.00	0.00	0.00	

07/29/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	8.61	0.00	0.00	0.00	3,226,003.00 *gal*
Application event totals		8.61	0.00	0.00	0.00	

H1 - 05/26/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/10/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	9.19	0.00	0.00	0.00	3,441,080.00 *gal*
Application event totals		9.19	0.00	0.00	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/22/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	10.57	0.00	0.00	0.00	3,957,234.00 *gal*
Application event totals		10.57	0.00	0.00	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/02/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	10.57	0.00	0.00	0.00	3,957,237.00 *gal*
Application event totals		10.57	0.00	0.00	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/14/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	10.57	0.00	0.00	0.00	3,957,234.00 *gal*
Application event totals		10.57	0.00	0.00	0.00	

M1 - 09/12/2022: Oats, silage-soft dough

Field name: M1

Crop: Oats, silage-soft dough

Plant date: 09/12/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/06/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	99.89	10.07	167.30	754.50	589,655.00 *gal*
Canal	Surface water	15.47	0.00	0.00	0.00	8,688,124.00 *gal*
Application event totals		115.36	10.07	167.30	754.50	

M1 - 09/12/2022: Oats, silage-soft dough

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/12/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	101.99	12.45	131.74	688.89	434,483.00 *gal*
Canal	Surface water	4.81	0.00	0.00	0.00	2,700,171.00 *gal*
Application event totals		106.80	12.45	131.74	688.89	

M1 - 05/25/2023: Corn, silage

Field name: M1

Crop: Corn, silage

Plant date: 05/25/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/21/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	9.14	0.00	0.00	0.00	5,132,296.00 *gal*
Application event totals		9.14	0.00	0.00	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/25/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	94.71	11.56	136.92	639.69	403,448.00 *gal*
Canal	Surface water	10.71	0.00	0.00	0.00	6,014,010.00 *gal*
Application event totals		105.41	11.56	136.92	639.69	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/08/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	8.74	0.00	0.00	0.00	4,909,403.00 *gal*
Application event totals		8.74	0.00	0.00	0.00	

M1 - 05/25/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/21/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	69.36	10.18	80.03	521.85	465,517.00 *gal*
Canal	Surface water	9.83	0.00	0.00	0.00	5,523,069.00 *gal*
Application event totals		79.19	10.18	80.03	521.85	

07/31/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	9.18	0.00	0.00	0.00	5,154,873.00 *gal*
Application event totals		9.18	0.00	0.00	0.00	

08/12/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	8.34	0.00	0.00	0.00	4,683,571.00 *gal*
Application event totals		8.34	0.00	0.00	0.00	

08/23/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	8.30	0.00	0.00	0.00	4,663,932.00 *gal*
Application event totals		8.30	0.00	0.00	0.00	

09/02/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	78.60	11.54	90.70	591.43	527,586.00 *gal*
Canal	Surface water	8.74	0.00	0.00	0.00	4,909,403.00 *gal*
Application event totals		87.34	11.54	90.70	591.43	

09/16/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	5.24	0.00	0.00	0.00	2,945,642.00 *gal*
Application event totals		5.24	0.00	0.00	0.00	

M2 - 09/12/2022: Oats, silage-soft dough

Field name: M2

Crop: Oats, silage-soft dough

Plant date: 09/12/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/06/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	99.89	10.07	167.30	754.50	550,345.00 *gal*
Canal	Surface water	15.47	0.00	0.00	0.00	8,108,915.00 *gal*
Application event totals		115.36	10.07	167.30	754.50	

04/12/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	101.99	12.45	147.45	688.89	405,517.00 *gal*
Canal	Surface water	4.81	0.00	0.00	0.00	2,520,160.00 *gal*
Application event totals		106.80	12.45	147.45	688.89	

M2 - 05/25/2023: Corn, silage

Field name: M2

Crop: Corn, silage

Plant date: 05/25/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/21/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	9.14	0.00	0.00	0.00	4,790,143.00 *gal*
Application event totals		9.14	0.00	0.00	0.00	

06/25/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	94.71	11.56	136.92	639.69	376,552.00 *gal*
Canal	Surface water	10.71	0.00	0.00	0.00	5,613,076.00 *gal*
Application event totals		105.41	11.56	136.92	639.69	

M2 - 05/25/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/08/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	8.74	0.00	0.00	0.00	4,582,109.00 *gal*
Application event totals		8.74	0.00	0.00	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/21/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	69.36	10.18	80.03	521.85	434,483.00 *gal*
Canal	Surface water	9.83	0.00	0.00	0.00	5,154,865.00 *gal*
Application event totals		79.19	10.18	80.03	521.85	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/31/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	9.18	0.00	0.00	0.00	4,811,215.00 *gal*
Application event totals		9.18	0.00	0.00	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/12/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	8.34	0.00	0.00	0.00	4,371,333.00 *gal*
Application event totals		8.34	0.00	0.00	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/23/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	8.30	0.00	0.00	0.00	4,353,004.00 *gal*
Application event totals		8.30	0.00	0.00	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/02/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	78.60	11.54	90.70	591.43	492,414.00 *gal*
Canal	Surface water	8.74	0.00	0.00	0.00	4,582,109.00 *gal*
Application event totals		87.34	11.54	90.70	591.43	

M2 - 05/25/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/16/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	5.24	0.00	0.00	0.00	2,749,265.00 *gal*
Application event totals		5.24	0.00	0.00	0.00	

N1 - 01/12/2016: Almond, in shell

Field name: N1

Crop: Almond, in shell

Plant date: 01/12/2016

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/23/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	14.76	0.00	0.00	0.00	4,420,801.00 *gal*
Application event totals		14.76	0.00	0.00	0.00	

05/10/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	2.29	0.00	0.00	0.00	686,802.00 *gal*
Application event totals		2.29	0.00	0.00	0.00	

05/22/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	2.11	0.00	0.00	0.00	631,855.00 *gal*
Application event totals		2.11	0.00	0.00	0.00	

06/05/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	2.20	0.00	0.00	0.00	659,328.00 *gal*
Application event totals		2.20	0.00	0.00	0.00	

N1 - 01/12/2016: Almond, in shell

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/20/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	2.66	0.00	0.00	0.00	796,695.00 *gal*
Application event totals		2.66	0.00	0.00	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/07/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	4.40	0.00	0.00	0.00	1,318,689.00 *gal*
Application event totals		4.40	0.00	0.00	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/20/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	2.20	0.00	0.00	0.00	659,328.00 *gal*
Application event totals		2.20	0.00	0.00	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/02/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	2.20	0.00	0.00	0.00	659,328.00 *gal*
Application event totals		2.20	0.00	0.00	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/11/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	2.20	0.00	0.00	0.00	659,328.00 *gal*
Application event totals		2.20	0.00	0.00	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/23/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	2.11	0.00	0.00	0.00	631,855.00 *gal*
Application event totals		2.11	0.00	0.00	0.00	

N1 - 01/12/2016: Almond, in shell

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/16/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.55	0.00	0.00	0.00	164,840.00 *gal*
Application event totals		0.55	0.00	0.00	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/23/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	4.40	0.00	0.00	0.00	1,318,689.00 *gal*
Application event totals		4.40	0.00	0.00	0.00	

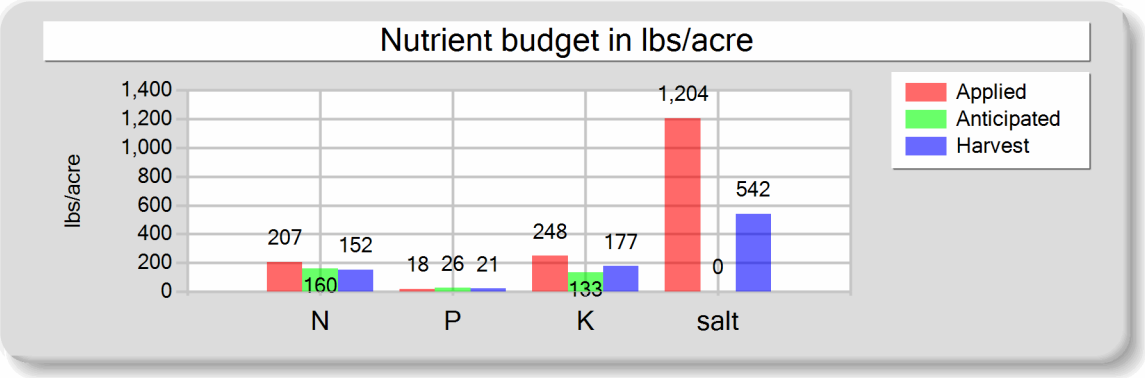
Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/19/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	4.31	0.00	0.00	0.00	1,291,216.00 *gal*
Application event totals		4.31	0.00	0.00	0.00	

B. NUTRIENT BUDGET

1 - 09/10/2022: Oats, silage-soft dough

Field name: 1 Crop: Oats, silage-soft dough Plant date: 09/10/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	180.57	18.24	247.99	1,204.46
Fresh water	19.68	0.00	0.00	0.00
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	207.25	18.24	247.99	1,204.46
Anticipated crop nutrient removal	160.00	25.60	132.80	0.00
Actual crop nutrient removal	152.10	20.89	176.91	541.82
Nutrient balance	55.15	-2.65	71.08	662.64
Applied to removed ratio	1.36	0.87	1.40	2.22

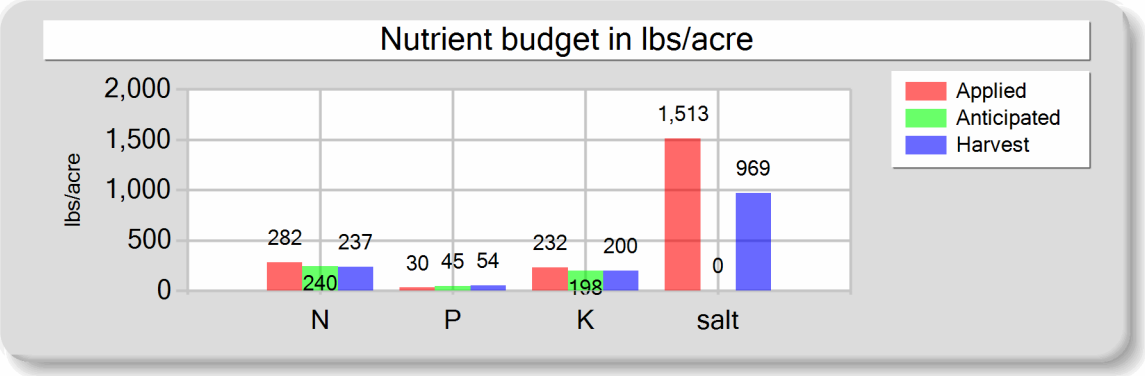
Fresh water applied	
3,315,844.00	*gallons*
122.11	*acre-inches*
13.57	*inches/acre*

Process wastewater applied	
255,000.00	*gallons*
9.39	*acre-inches*
1.04	*inches/acre*

Total harvests for the crop	
1	*harvests*

1 - 05/22/2023: Corn, silage

Field name: 1 Crop: Corn, silage Plant date: 05/22/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	201.13	29.52	232.07	1,513.37
Fresh water	73.38	0.00	0.00	0.00
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	281.52	29.52	232.07	1,513.37
Anticipated crop nutrient removal	240.00	45.00	198.00	0.00
Actual crop nutrient removal	236.66	54.00	200.13	968.90
Nutrient balance	44.85	-24.49	31.94	544.47
Applied to removed ratio	1.19	0.55	1.16	1.56

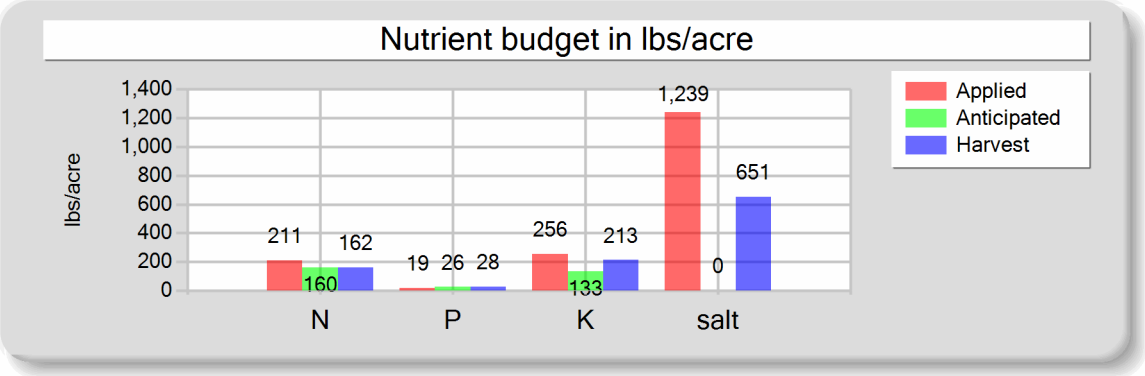
Fresh water applied	
12,366,240.00	*gallons*
455.41	*acre-inches*
50.60	*inches/acre*

Process wastewater applied	
405,000.00	*gallons*
14.91	*acre-inches*
1.66	*inches/acre*

Total harvests for the crop	
1	*harvests*

2 - 09/10/2022: Oats, silage-soft dough

Field name: 2 Crop: Oats, silage-soft dough Plant date: 09/10/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	184.66	18.65	255.95	1,238.64
Fresh water	19.68	0.00	0.00	0.00
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	211.34	18.65	255.95	1,238.64
Anticipated crop nutrient removal	160.00	25.60	132.80	0.00
Actual crop nutrient removal	162.06	27.90	212.54	650.91
Nutrient balance	49.28	-9.25	43.41	587.73
Applied to removed ratio	1.30	0.67	1.20	1.90

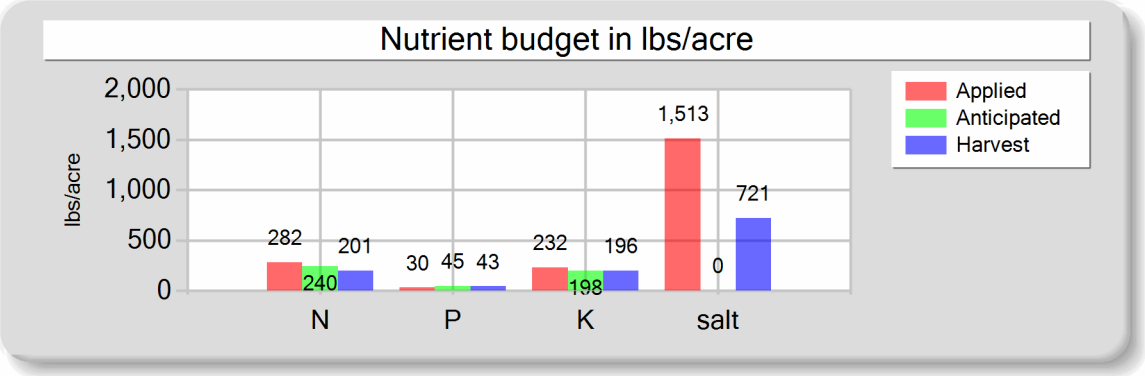
Fresh water applied	
7,736,971.00	*gallons*
284.93	*acre-inches*
13.57	*inches/acre*

Process wastewater applied	
615,000.00	*gallons*
22.65	*acre-inches*
1.08	*inches/acre*

Total harvests for the crop	
1	*harvests*

2 - 05/22/2023: Corn, silage

Field name: 2 Crop: Corn, silage Plant date: 05/22/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	201.13	29.52	232.07	1,513.37
Fresh water	73.38	0.00	0.00	0.00
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	281.52	29.52	232.07	1,513.37
Anticipated crop nutrient removal	240.00	45.00	198.00	0.00
Actual crop nutrient removal	201.09	42.98	196.48	721.46
Nutrient balance	80.43	-13.46	35.59	791.91
Applied to removed ratio	1.40	0.69	1.18	2.10

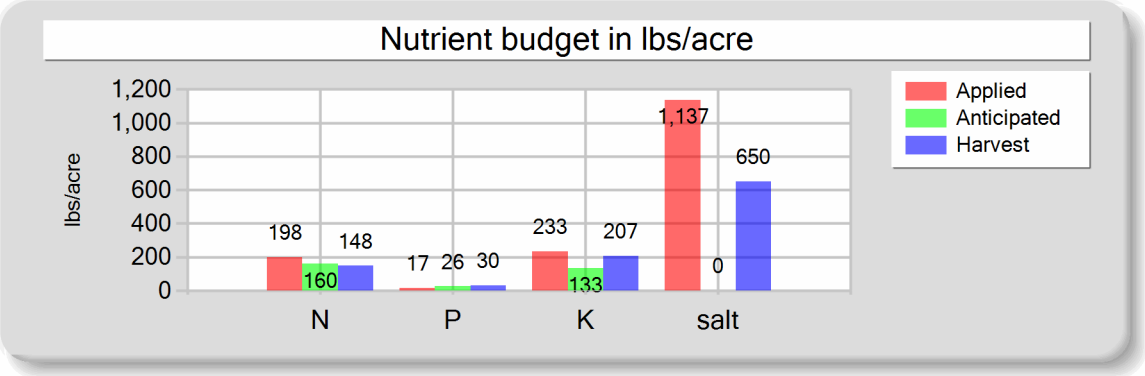
Fresh water applied	
28,854,559.00	*gallons*
1,062.62	*acre-inches*
50.60	*inches/acre*

Process wastewater applied	
945,000.00	*gallons*
34.80	*acre-inches*
1.66	*inches/acre*

Total harvests for the crop	
1	*harvests*

3 - 09/10/2022: Oats, silage-soft dough

Field name: 3 Crop: Oats, silage-soft dough Plant date: 09/10/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	171.48	17.32	233.11	1,136.82
Fresh water	19.68	0.00	0.00	0.00
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	198.15	17.32	233.11	1,136.82
Anticipated crop nutrient removal	160.00	25.60	132.80	0.00
Actual crop nutrient removal	147.61	30.37	206.93	649.75
Nutrient balance	50.55	-13.05	26.18	487.06
Applied to removed ratio	1.34	0.57	1.13	1.75

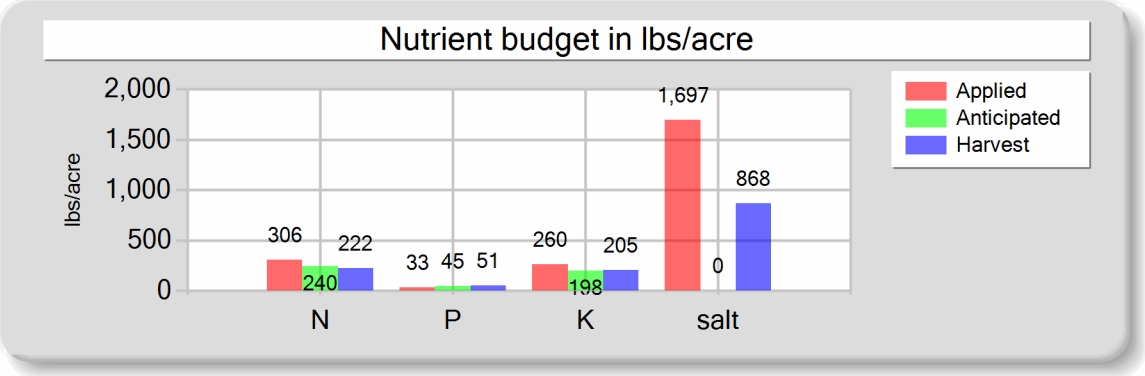
Fresh water applied	
8,105,398.00	*gallons*
298.49	*acre-inches*
13.57	*inches/acre*

Process wastewater applied	
585,000.00	*gallons*
21.54	*acre-inches*
0.98	*inches/acre*

Total harvests for the crop	
1	*harvests*

3 - 05/22/2023: Corn, silage

Field name: 3 Crop: Corn, silage Plant date: 05/22/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	225.51	33.09	260.20	1,696.80
Fresh water	73.38	0.00	0.00	0.00
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	305.89	33.09	260.20	1,696.80
Anticipated crop nutrient removal	240.00	45.00	198.00	0.00
Actual crop nutrient removal	221.65	51.15	204.60	867.99
Nutrient balance	84.25	-18.06	55.61	828.81
Applied to removed ratio	1.38	0.65	1.27	1.95

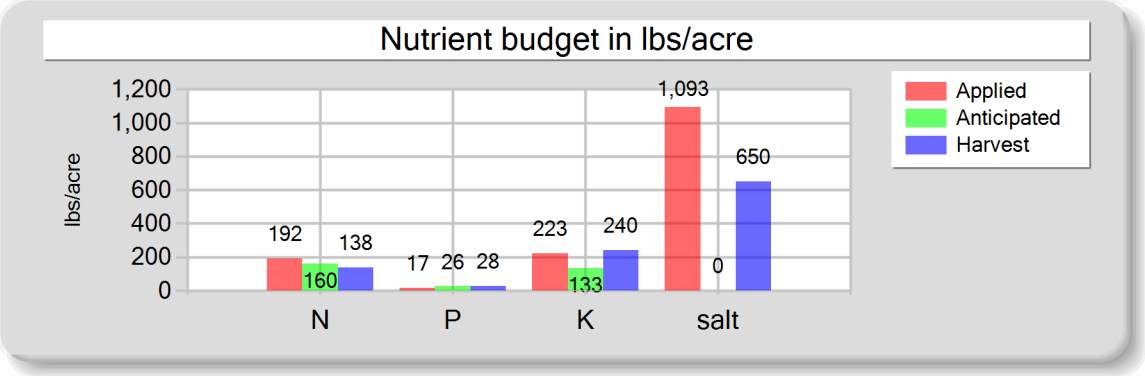
Fresh water applied	
30,228,587.00	*gallons*
1,113.22	*acre-inches*
50.60	*inches/acre*

Process wastewater applied	
1,110,000.00	*gallons*
40.88	*acre-inches*
1.86	*inches/acre*

Total harvests for the crop	
1	*harvests*

4 - 09/10/2022: Oats, silage-soft dough

Field name: 4 Crop: Oats, silage-soft dough Plant date: 09/10/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	165.51	16.72	223.48	1,092.81
Fresh water	19.68	0.00	0.00	0.00
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	192.18	16.72	223.48	1,092.81
Anticipated crop nutrient removal	160.00	25.60	132.80	0.00
Actual crop nutrient removal	138.03	28.50	240.43	650.06
Nutrient balance	54.16	-11.78	-16.95	442.75
Applied to removed ratio	1.39	0.59	0.93	1.68

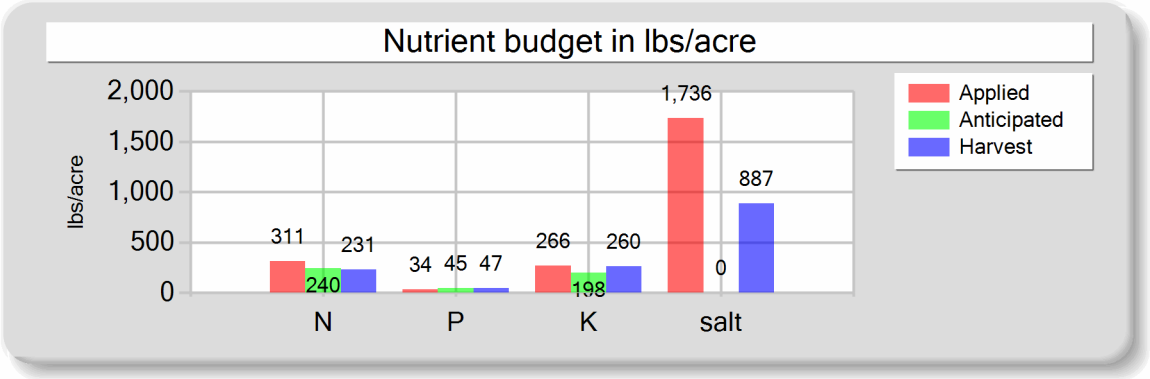
Fresh water applied	
15,842,367.00	*gallons*
583.42	*acre-inches*
13.57	*inches/acre*

Process wastewater applied	
1,095,000.00	*gallons*
40.33	*acre-inches*
0.94	*inches/acre*

Total harvests for the crop	
1	*harvests*

4 - 05/22/2023: Corn, silage

Field name: 4 Crop: Corn, silage Plant date: 05/22/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	230.76	33.86	266.26	1,736.26
Fresh water	73.38	0.00	0.00	0.00
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	311.14	33.86	266.26	1,736.26
Anticipated crop nutrient removal	240.00	45.00	198.00	0.00
Actual crop nutrient removal	231.28	46.55	260.37	887.29
Nutrient balance	79.86	-12.68	5.89	848.98
Applied to removed ratio	1.35	0.73	1.02	1.96

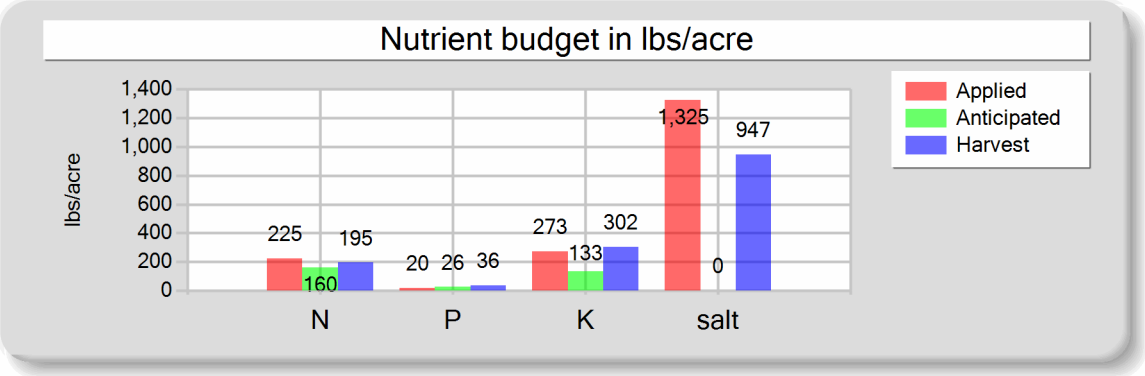
Fresh water applied	
59,083,148.00	*gallons*
2,175.83	*acre-inches*
50.60	*inches/acre*

Process wastewater applied	
2,220,000.00	*gallons*
81.76	*acre-inches*
1.90	*inches/acre*

Total harvests for the crop	
1	*harvests*

5 - 09/10/2022: Oats, silage-soft dough

Field name: 5 Crop: Oats, silage-soft dough Plant date: 09/10/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	198.01	20.00	273.30	1,324.77
Fresh water	19.68	0.00	0.00	0.00
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	224.68	20.00	273.30	1,324.77
Anticipated crop nutrient removal	160.00	25.60	132.80	0.00
Actual crop nutrient removal	194.96	36.44	302.46	947.47
Nutrient balance	29.72	-16.44	-29.16	377.29
Applied to removed ratio	1.15	0.55	0.90	1.40

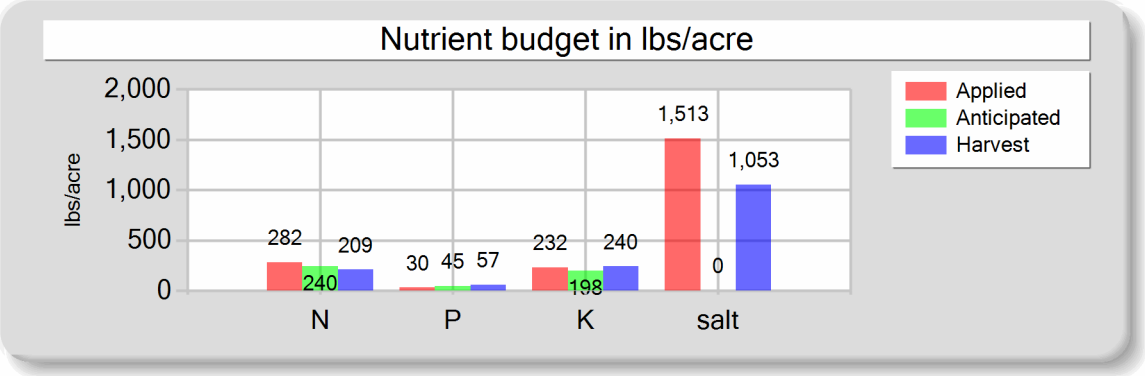
Fresh water applied	
4,421,125.00	*gallons*
162.81	*acre-inches*
13.57	*inches/acre*

Process wastewater applied	
375,000.00	*gallons*
13.81	*acre-inches*
1.15	*inches/acre*

Total harvests for the crop	
1	*harvests*

5 - 05/22/2023: Corn, silage

Field name: 5 Crop: Corn, silage Plant date: 05/22/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	201.13	29.52	232.07	1,513.37
Fresh water	73.38	0.00	0.00	0.00
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	281.52	29.52	232.07	1,513.37
Anticipated crop nutrient removal	240.00	45.00	198.00	0.00
Actual crop nutrient removal	208.80	57.28	240.21	1,053.22
Nutrient balance	72.72	-27.76	-8.13	460.15
Applied to removed ratio	1.35	0.52	0.97	1.44

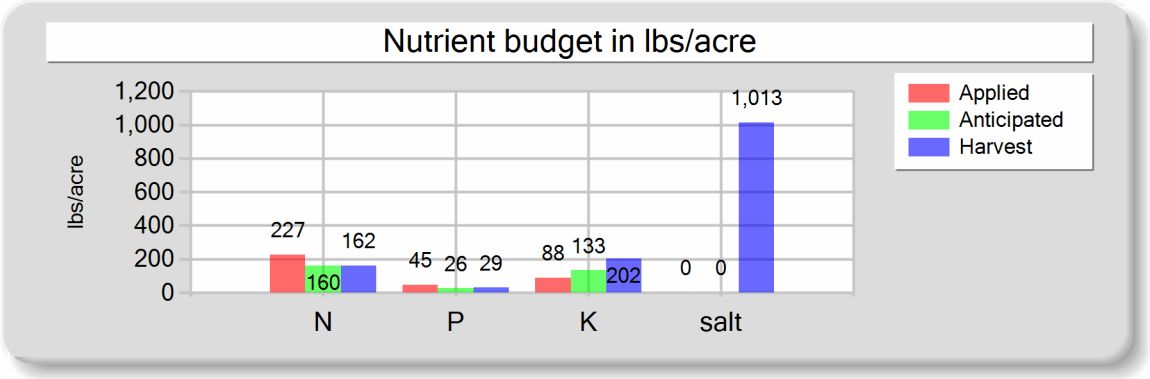
Fresh water applied	
16,488,319.00	*gallons*
607.21	*acre-inches*
50.60	*inches/acre*

Process wastewater applied	
540,000.00	*gallons*
19.89	*acre-inches*
1.66	*inches/acre*

Total harvests for the crop	
1	*harvests*

C1 - 09/15/2022: Oats, silage-soft dough

Field name: C1 Crop: Oats, silage-soft dough Plant date: 09/15/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	199.67	45.07	88.26	0.00
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	20.56	0.00	0.00	0.00
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	227.23	45.07	88.26	0.00
Anticipated crop nutrient removal	160.00	25.60	132.80	0.00
Actual crop nutrient removal	162.13	29.22	201.72	1,013.34
Nutrient balance	65.10	15.84	-113.47	-1,013.34
Applied to removed ratio	1.40	1.54	0.44	0.00

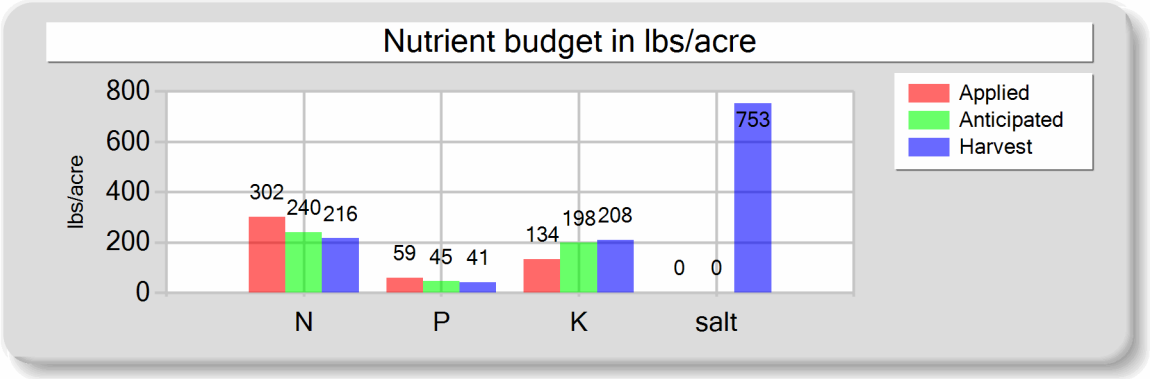
Fresh water applied	
11,934,847.00	*gallons*
439.52	*acre-inches*
14.18	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
1	*harvests*

C1 - 05/26/2023: Corn, silage

Field name: C1 Crop: Corn, silage Plant date: 05/26/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	214.63	59.10	133.76	0.00
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	80.52	0.00	0.00	0.00
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	302.15	59.10	133.76	0.00
Anticipated crop nutrient removal	240.00	45.00	198.00	0.00
Actual crop nutrient removal	215.72	40.54	208.48	752.85
Nutrient balance	86.43	18.56	-74.73	-752.85
Applied to removed ratio	1.40	1.46	0.64	0.00

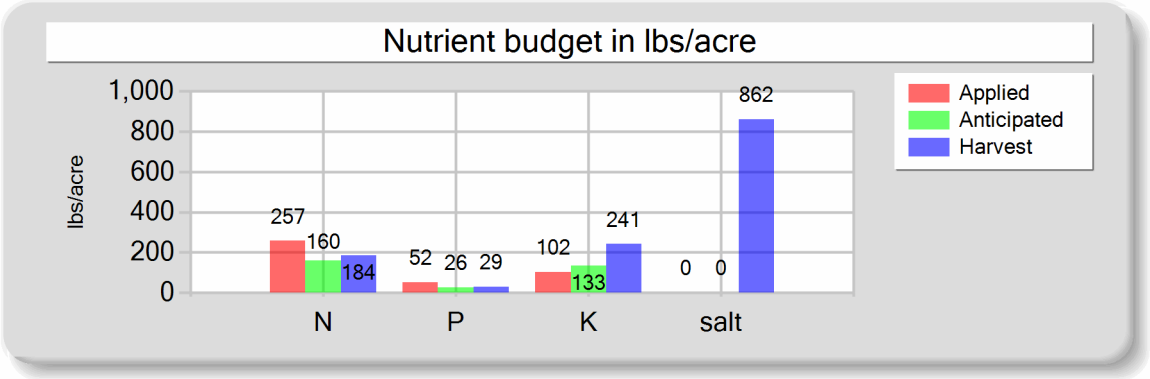
Fresh water applied	
46,738,190.00	*gallons*
1,721.21	*acre-inches*
55.52	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
1	*harvests*

C2 - 09/15/2022: Oats, silage-soft dough

Field name: C2 Crop: Oats, silage-soft dough Plant date: 09/15/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	229.83	51.87	101.59	0.00
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	20.56	0.00	0.00	0.00
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	257.39	51.87	101.59	0.00
Anticipated crop nutrient removal	160.00	25.60	132.80	0.00
Actual crop nutrient removal	183.85	29.16	241.32	862.17
Nutrient balance	73.54	22.71	-139.74	-862.17
Applied to removed ratio	1.40	1.78	0.42	0.00

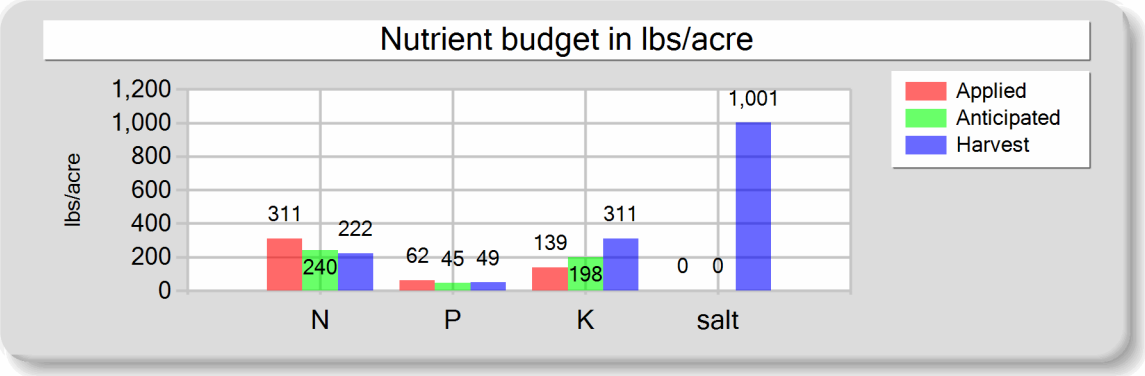
Fresh water applied	
15,784,797.00	*gallons*
581.30	*acre-inches*
14.18	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
1	*harvests*

C2 - 05/26/2023: Corn, silage

Field name: C2 Crop: Corn, silage Plant date: 05/26/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	223.80	61.63	139.47	0.00
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	80.52	0.00	0.00	0.00
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	311.33	61.63	139.47	0.00
Anticipated crop nutrient removal	240.00	45.00	198.00	0.00
Actual crop nutrient removal	222.26	48.57	310.58	1,000.93
Nutrient balance	89.06	13.05	-171.11	-1,000.93
Applied to removed ratio	1.40	1.27	0.45	0.00

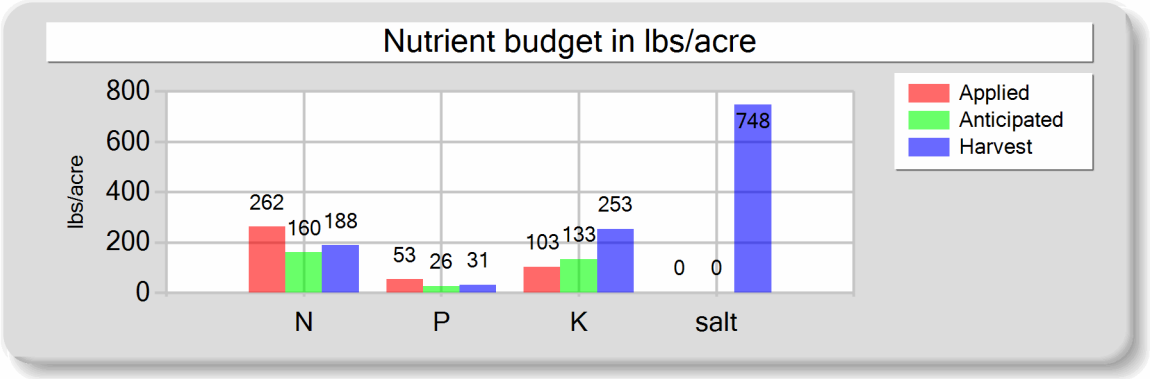
Fresh water applied	
61,815,025.00	*gallons*
2,276.44	*acre-inches*
55.52	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
1	*harvests*

H1 - 09/11/2022: Oats, silage-soft dough

Field name: H1 Crop: Oats, silage-soft dough Plant date: 09/11/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	233.41	52.68	103.17	0.00
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	21.21	0.00	0.00	0.00
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	261.62	52.68	103.17	0.00
Anticipated crop nutrient removal	160.00	25.60	132.80	0.00
Actual crop nutrient removal	187.75	31.42	252.83	748.02
Nutrient balance	73.87	21.27	-149.66	-748.02
Applied to removed ratio	1.39	1.68	0.41	0.00

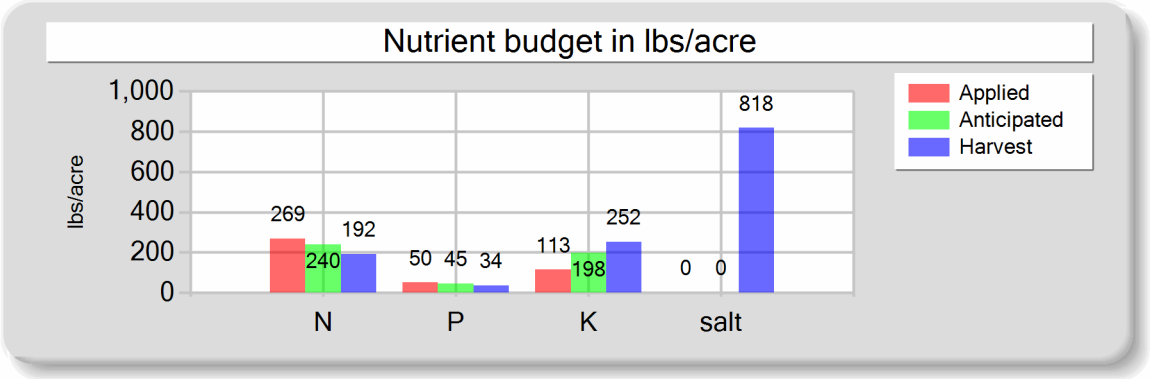
Fresh water applied	
7,942,761.00	*gallons*
292.50	*acre-inches*
14.63	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
1	*harvests*

H1 - 05/26/2023: Corn, silage

Field name: H1 Crop: Corn, silage Plant date: 05/26/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	182.02	50.12	113.43	0.00
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	80.13	0.00	0.00	0.00
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	269.15	50.12	113.43	0.00
Anticipated crop nutrient removal	240.00	45.00	198.00	0.00
Actual crop nutrient removal	192.03	34.38	252.48	817.90
Nutrient balance	77.13	15.75	-139.05	-817.90
Applied to removed ratio	1.40	1.46	0.45	0.00

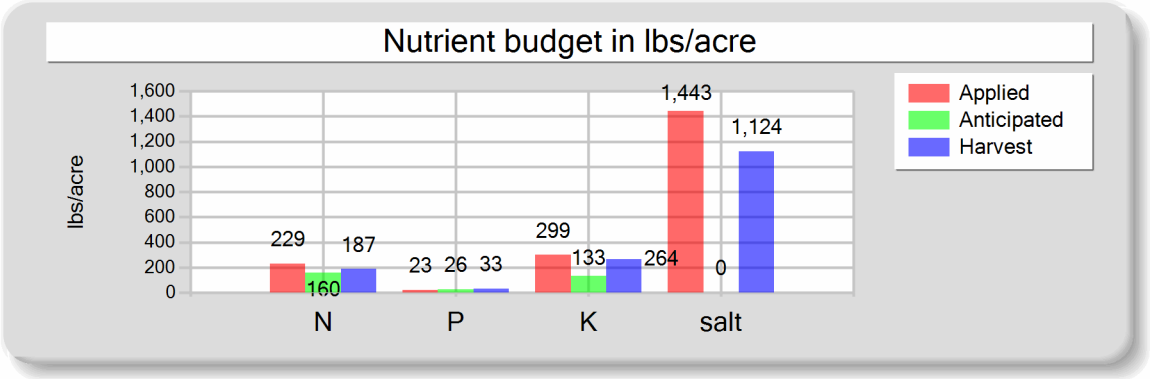
Fresh water applied	
30,008,317.00	*gallons*
1,105.10	*acre-inches*
55.26	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
1	*harvests*

M1 - 09/12/2022: Oats, silage-soft dough

Field name: M1 Crop: Oats, silage-soft dough Plant date: 09/12/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	201.88	22.52	299.04	1,443.40
Fresh water	20.27	0.00	0.00	0.00
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	229.16	22.52	299.04	1,443.40
Anticipated crop nutrient removal	160.00	25.60	132.80	0.00
Actual crop nutrient removal	187.19	32.64	264.05	1,123.96
Nutrient balance	41.97	-10.12	34.99	319.44
Applied to removed ratio	1.22	0.69	1.13	1.28

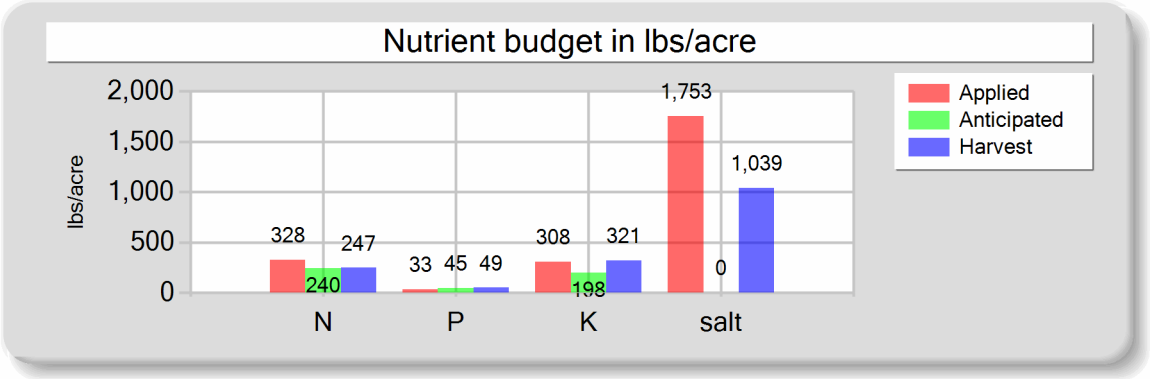
Fresh water applied	
11,388,295.00	*gallons*
419.39	*acre-inches*
13.98	*inches/acre*

Process wastewater applied	
1,024,138.00	*gallons*
37.72	*acre-inches*
1.26	*inches/acre*

Total harvests for the crop	
1	*harvests*

M1 - 05/25/2023: Corn, silage

Field name: M1 Crop: Corn, silage Plant date: 05/25/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	242.67	33.27	307.64	1,752.97
Fresh water	78.22	0.00	0.00	0.00
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	327.88	33.27	307.64	1,752.97
Anticipated crop nutrient removal	240.00	45.00	198.00	0.00
Actual crop nutrient removal	246.69	48.69	321.34	1,038.68
Nutrient balance	81.20	-15.42	-13.70	714.29
Applied to removed ratio	1.33	0.68	0.96	1.69

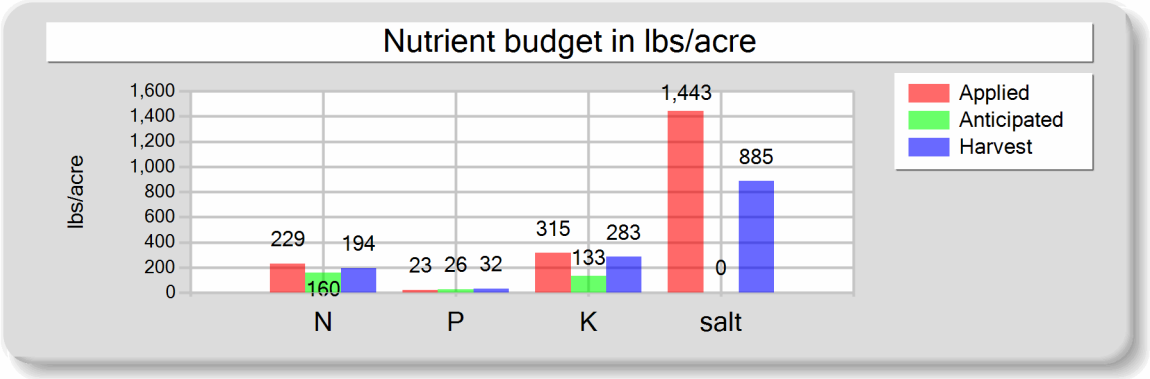
Fresh water applied	
43,936,199.00	*gallons*
1,618.02	*acre-inches*
53.93	*inches/acre*

Process wastewater applied	
1,396,551.00	*gallons*
51.43	*acre-inches*
1.71	*inches/acre*

Total harvests for the crop	
1	*harvests*

M2 - 09/12/2022: Oats, silage-soft dough

Field name: M2 Crop: Oats, silage-soft dough Plant date: 09/12/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	201.88	22.52	314.75	1,443.40
Fresh water	20.27	0.00	0.00	0.00
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	229.16	22.52	314.75	1,443.40
Anticipated crop nutrient removal	160.00	25.60	132.80	0.00
Actual crop nutrient removal	194.41	31.85	283.34	885.18
Nutrient balance	34.75	-9.33	31.41	558.22
Applied to removed ratio	1.18	0.71	1.11	1.63

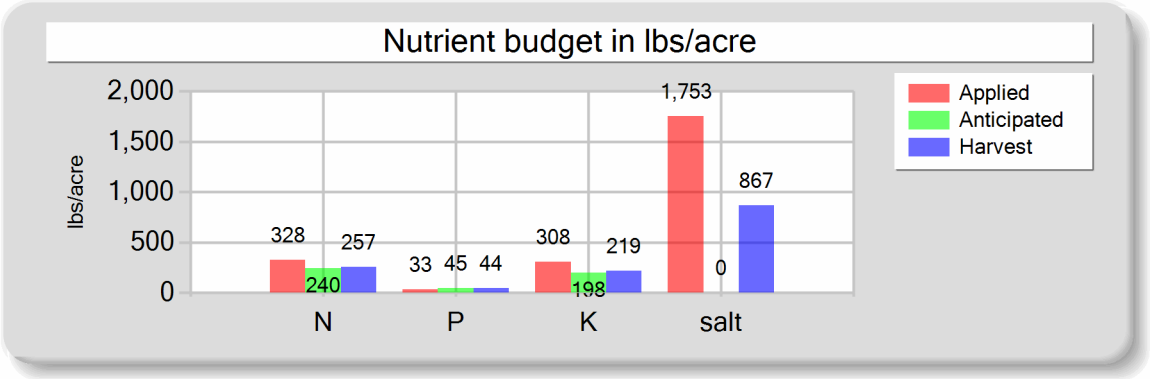
Fresh water applied	
10,629,075.00	*gallons*
391.43	*acre-inches*
13.98	*inches/acre*

Process wastewater applied	
955,862.00	*gallons*
35.20	*acre-inches*
1.26	*inches/acre*

Total harvests for the crop	
1	*harvests*

M2 - 05/25/2023: Corn, silage

Field name: M2 Crop: Corn, silage Plant date: 05/25/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	242.67	33.27	307.64	1,752.97
Fresh water	78.22	0.00	0.00	0.00
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	327.88	33.27	307.64	1,752.97
Anticipated crop nutrient removal	240.00	45.00	198.00	0.00
Actual crop nutrient removal	257.18	44.09	218.97	867.06
Nutrient balance	70.70	-10.82	88.67	885.90
Applied to removed ratio	1.27	0.75	1.40	2.02

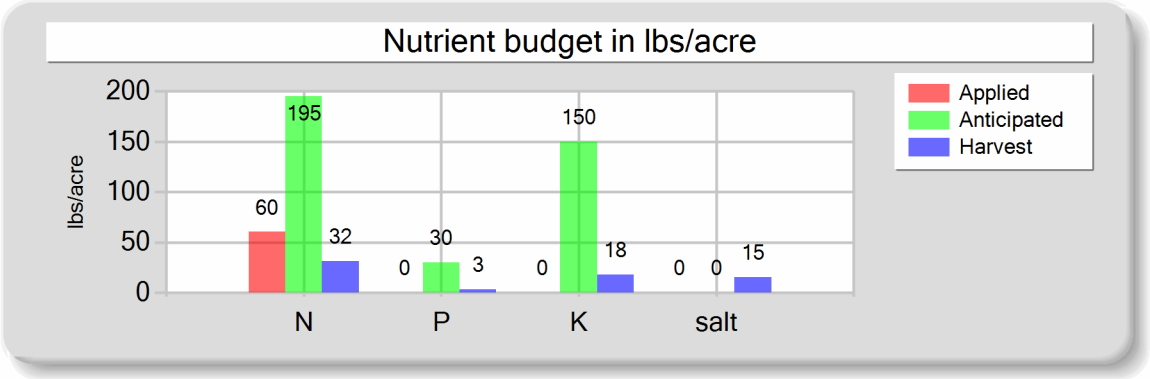
Fresh water applied	
41,007,119.00	*gallons*
1,510.15	*acre-inches*
53.93	*inches/acre*

Process wastewater applied	
1,303,449.00	*gallons*
48.00	*acre-inches*
1.71	*inches/acre*

Total harvests for the crop	
1	*harvests*

N1 - 01/12/2016: Almond, in shell

Field name: N1 Crop: Almond, in shell Plant date: 01/12/2016



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	46.39	0.00	0.00	0.00
Atmospheric deposition	14.00	0.00	0.00	0.00
Total nutrients applied	60.39	0.00	0.00	0.00
Anticipated crop nutrient removal	195.00	30.00	150.00	0.00
Actual crop nutrient removal	31.50	3.36	17.82	15.36
Nutrient balance	28.89	-3.36	-17.82	-15.36
Applied to removed ratio	1.92	0.00	0.00	0.00

Fresh water applied	
13,898,754.00	*gallons*
511.84	*acre-inches*
31.99	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
1	*harvests*

NUTRIENT ANALYSES

A. MANURE ANALYSES

Separator Manure - 2022

Sample and source description: Separator Manure - 2022

Sample date: 10/20/2022 Material type: Separator solids Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 72.9 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Calcium (mg/kg)	Magnesium (mg/kg)	Sodium (mg/kg)	Sulfur (mg/kg)	Chloride (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	31,900.00	7,200.00	14,100.00							0.00
DL	100.00	100.00	100.00							1.00

Separator Manure

Sample and source description: Separator Manure

Sample date: 05/08/2023 Material type: Separator solids Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 75.3 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Calcium (mg/kg)	Magnesium (mg/kg)	Sodium (mg/kg)	Sulfur (mg/kg)	Chloride (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	27,600.00	7,600.00	17,200.00							0.00
DL	100.00	100.00	100.00							1.00

Separator Manure

Sample and source description: Separator Manure

Sample date: 11/14/2023 Material type: Separator solids Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 73.9 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Calcium (mg/kg)	Magnesium (mg/kg)	Sodium (mg/kg)	Sulfur (mg/kg)	Chloride (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	17,600.00	3,200.00	4,200.00							0.00
DL	100.00	100.00	100.00							1.00

B. PROCESS WASTEWATER ANALYSES

Lagoon Q4 - 2022

Sample and source description: Lagoon Q4 - 2022

Sample date: 11/02/2022 Material type: Process wastewater Source of analysis: Lab analysis pH: 0.00

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	607.00	381.00	0.00	2.00	61.40	1,020.00								9,600.00	4,600
DL	0.01	0.01	0.01	0.01	0.01	0.01								0.01	1

Lagoon Q1

Sample and source description: Lagoon Q1

Sample date: 03/30/2023 Material type: Process wastewater Source of analysis: Lab analysis pH: 7.50

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	983.00	636.00	0.00	1.70	99.60	1,090.00	365.00	167.00	238.00	5,120.00	0.00	5.20	520.00	11,000.00	5,800
DL	0.01	0.01	0.01	0.01	0.01	0.01	0.01	0.01	0.01	0.01	0.01	0.01	0.01	0.01	1

Lagoon Q2

Sample and source description: Lagoon Q2

Sample date: 05/08/2023 Material type: Process wastewater Source of analysis: Lab analysis pH:

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	843.00	465.00	0.00	0.90	103.00	1,220.00								11,600.00	5,700
DL	0.01	0.01	0.01	0.01	0.01	0.01								0.01	1

Lagoon Q3

Sample and source description: Lagoon Q3

Sample date: 08/02/2023 Material type: Process wastewater Source of analysis: Lab analysis pH: 7.40

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	535.00	485.00	0.00	0.60	78.60	618.00								8,390.00	4,030
DL	0.01	0.01	0.01	0.01	0.01	0.01								0.01	1

Lagoon Q4

Sample and source description: Lagoon Q4

Sample date: 11/14/2023 Material type: Process wastewater Source of analysis: Lab analysis pH: 7.60

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	629.00	449.00	0.00	0.90	75.20	701.00								5,530.00	4,200
DL	0.01	0.01	0.01	0.01	0.01	0.01								0.01	1

C. FRESH WATER ANALYSES

Canal

TID Canal

Sample description: TID Canal

Sample date: 09/20/2023 Source of analysis: Lab analysis

	Total N (mg/L)	NH4-N (mg/L)	Nitrate-N (mg/L)	Calcium (mg/L)	Magnesium (mg/L)	Sodium (mg/L)	Bicarbonate (mg/L)	Carbonate (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value		0.00	6.40								284.00	0
DL		0.01	0.01								0.01	1

Domestic Well

DW - MB

Sample description: DW - MB

Sample date: 12/06/2023 Source of analysis: Lab analysis

	Total N (mg/L)	NH4-N (mg/L)	Nitrate-N (mg/L)	Calcium (mg/L)	Magnesium (mg/L)	Sodium (mg/L)	Bicarbonate (mg/L)	Carbonate (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value		0.00	31.10								794.00	0
DL		0.01	0.01								0.01	1

Domestic Well

DW - Office

Sample description: DW - Office

Sample date: 12/06/2023 Source of analysis: Lab analysis

	Total N (mg/L)	NH4-N (mg/L)	Nitrate-N (mg/L)	Calcium (mg/L)	Magnesium (mg/L)	Sodium (mg/L)	Bicarbonate (mg/L)	Carbonate (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value		0.00	26.60								728.00	0
DL		0.01	0.01								0.01	1

D. SOIL ANALYSES

No soil analyses entered.

E. PLANT TISSUE ANALYSES

1 - 09/10/2022: Oats, silage-soft dough

Field 1 Oat Silage

Sample and source description: Field 1 Oat Silage

Sample date: 05/10/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 78.4 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	13,300.00	3,200.00	27,100.00		8.30
DL	100.00	100.00	100.00		1.00

1 - 05/22/2023: Corn, silage

1 - 05/22/2023: Corn, silage

Corn Silage Field 1

Sample and source description: Corn Silage Field 1

Sample date: 09/28/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 66.6 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	14,900.00	3,400.00	12,600.00		6.10
DL	100.00	100.00	100.00		1.00

2 - 09/10/2022: Oats, silage-soft dough

Field 2 Oat Silage

Sample and source description: Field 2 Oat Silage

Sample date: 05/10/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 78.0 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	24,400.00	4,200.00	32,000.00		9.80
DL	100.00	100.00	100.00		1.00

2 - 05/22/2023: Corn, silage

Corn Silage Field 2

Sample and source description: Corn Silage Field 2

Sample date: 10/02/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 67.7 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	13,100.00	2,800.00	12,800.00		4.70
DL	100.00	100.00	100.00		1.00

3 - 09/10/2022: Oats, silage-soft dough

Field 3 Oat Silage

Sample and source description: Field 3 Oat Silage

Sample date: 05/10/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 76.6 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	20,900.00	4,300.00	29,300.00		9.20
DL	100.00	100.00	100.00		1.00

3 - 05/22/2023: Corn, silage

Corn Silage Field 3

Sample and source description: Corn Silage Field 3

Sample date: 10/02/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 67.4 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	14,300.00	3,300.00	13,200.00		5.60
DL	100.00	100.00	100.00		1.00

4 - 09/10/2022: Oats, silage-soft dough

Field 4 Oat Silage

Sample and source description: Field 4 Oat Silage

Sample date: 05/10/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 70.5 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	15,500.00	3,200.00	27,000.00		7.30
DL	100.00	100.00	100.00		1.00

4 - 05/22/2023: Corn, silage

Corn Silage Field 4

Sample and source description: Corn Silage Field 4

Sample date: 09/29/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 69.4 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	15,900.00	3,200.00	17,900.00		6.10
DL	100.00	100.00	100.00		1.00

5 - 09/10/2022: Oats, silage-soft dough

Field 5 Oat Silage

Sample and source description: Field 5 Oat Silage

Sample date: 05/09/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 69.8 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	21,400.00	4,000.00	33,200.00		10.40
DL	100.00	100.00	100.00		1.00

5 - 05/22/2023: Corn, silage

Corn Silage Field 5

Sample and source description: Corn Silage Field 5

Sample date: 09/28/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 61.1 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	11,300.00	3,100.00	13,000.00		5.70
DL	100.00	100.00	100.00		1.00

C1 - 09/15/2022: Oats, silage-soft dough

Field C1 Oat Silage

Sample and source description: Field C1 Oat Silage

Sample date: 05/08/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 64.7 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	13,600.00	3,300.00	19,100.00		12.20
DL	100.00	100.00	100.00		1.00

Field C1/C2 Oat Silage

Sample and source description: Field C1/C2 Oat Silage

Sample date: 05/09/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 67.4 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	20,800.00	2,900.00	23,700.00		9.30
DL	100.00	100.00	100.00		1.00

C1 - 05/26/2023: Corn, silage

Corn Silage Field C1

Sample and source description: Corn Silage Field C1

Sample date: 09/29/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 65.1 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	14,900.00	2,800.00	14,400.00		5.20
DL	100.00	100.00	100.00		1.00

C2 - 09/15/2022: Oats, silage-soft dough

C2 - 09/15/2022: Oats, silage-soft dough

Field C1/C2 Oat Silage

Sample and source description: Field C1/C2 Oat Silage

Sample date: 05/09/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 67.4 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	20,800.00	2,900.00	23,700.00		9.30
DL	100.00	100.00	100.00		1.00

Field C2 Oat Silage

Sample and source description: Field C2 Oat Silage

Sample date: 05/09/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 71.8 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	22,700.00	4,000.00	33,400.00		11.10
DL	100.00	100.00	100.00		1.00

C2 - 05/26/2023: Corn, silage

Corn Silage Field C2

Sample and source description: Corn Silage Field C2

Sample date: 09/29/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 64.5 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	15,100.00	3,300.00	21,100.00		6.80
DL	100.00	100.00	100.00		1.00

H1 - 09/11/2022: Oats, silage-soft dough

H1 - 09/11/2022: Oats, silage-soft dough

Field H1 Oat Silage

Sample and source description: Field H1 Oat Silage

Sample date: 05/08/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 72.7 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	25,100.00	4,200.00	33,800.00		10.00
DL	100.00	100.00	100.00		1.00

H1 - 05/26/2023: Corn, silage

Corn Silage Field H1

Sample and source description: Corn Silage Field H1

Sample date: 09/29/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 73.6 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	16,200.00	2,900.00	21,300.00		6.90
DL	100.00	100.00	100.00		1.00

M1 - 09/12/2022: Oats, silage-soft dough

Field M1 Oat Silage

Sample and source description: Field M1 Oat Silage

Sample date: 05/08/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 72.1 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	23,800.00	3,900.00	29,400.00		15.70
DL	100.00	100.00	100.00		1.00

M1 - 09/12/2022: Oats, silage-soft dough

Field M1/M2 Oat Silage

Sample and source description: Field M1/M2 Oat Silage

Sample date: 05/08/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 69.7 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	21,500.00	4,000.00	34,500.00		11.50
DL	100.00	100.00	100.00		1.00

M1 - 05/25/2023: Corn, silage

Corn Silage Field M1

Sample and source description: Corn Silage Field M1

Sample date: 09/29/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 66.0 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	15,200.00	3,000.00	19,800.00		6.40
DL	100.00	100.00	100.00		1.00

M2 - 09/12/2022: Oats, silage-soft dough

Field M1/M2 Oat Silage

Sample and source description: Field M1/M2 Oat Silage

Sample date: 05/08/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 69.7 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	21,500.00	4,000.00	34,500.00		11.50
DL	100.00	100.00	100.00		1.00

M2 - 09/12/2022: Oats, silage-soft dough

Field M2 Oat Silage

Sample and source description: Field M2 Oat Silage

Sample date: 05/08/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 72.1 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	25,500.00	3,700.00	34,000.00		9.90
DL	100.00	100.00	100.00		1.00

M2 - 05/25/2023: Corn, silage

Corn Silage Field M2

Sample and source description: Corn Silage Field M2

Sample date: 09/29/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 69.2 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	17,500.00	3,000.00	14,900.00		5.90
DL	100.00	100.00	100.00		1.00

F. SUBSURFACE (TILE) DRAINAGE ANALYSES

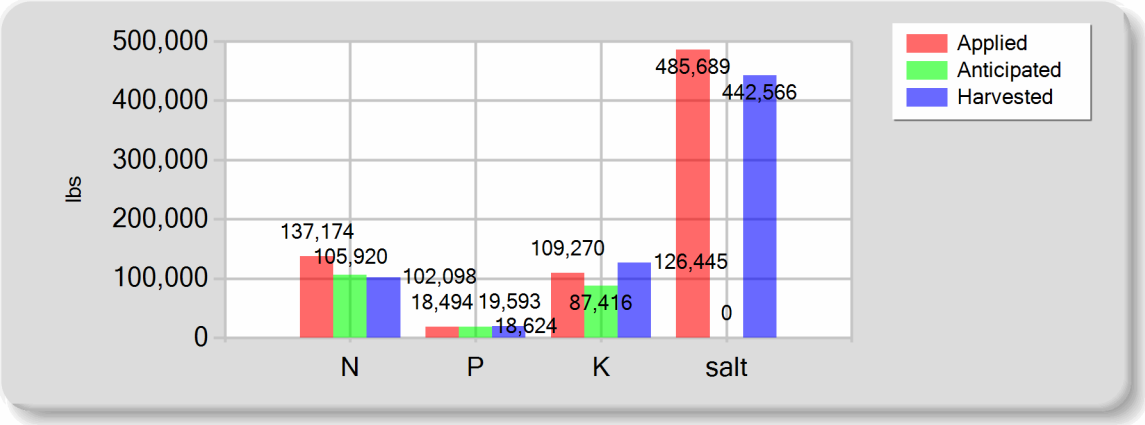
No subsurface (tile) drainage analyses entered.

NUTRIENT APPLICATIONS, POTENTIAL REMOVAL, AND BALANCE

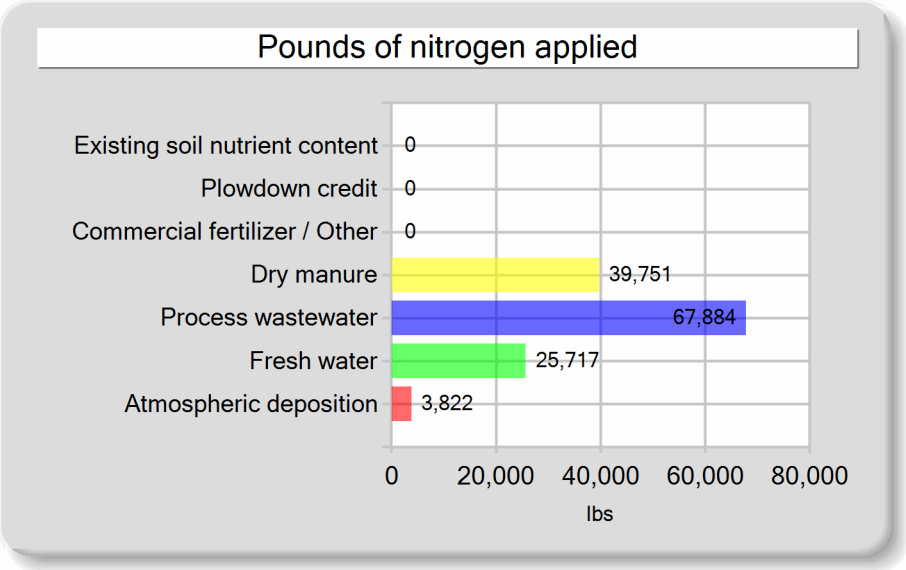
A. SUMMARY OF NUTRIENT APPLICATIONS, POTENTIAL REMOVAL, AND BALANCE

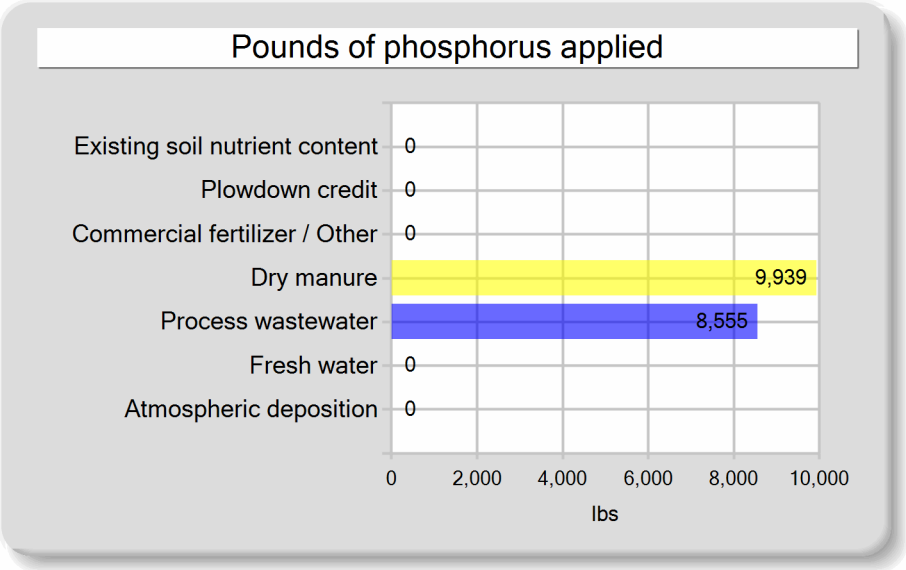
	Total N (lbs)	Total P (lbs)	Total K (lbs)	Total salt (lbs)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	39,750.86	9,938.80	21,097.73	0.00
Process wastewater	67,883.52	8,555.41	88,172.33	485,689.01
Fresh water	25,717.34	0.00	0.00	0.00
Atmospheric deposition	3,822.00	0.00	0.00	0.00
Total nutrients applied	137,173.71	18,494.21	109,270.06	485,689.01
Anticipated crop nutrient removal	105,920.00	18,624.20	87,415.60	0.00
Actual crop nutrient removal	102,098.13	19,592.84	126,445.16	442,565.57
Nutrient balance	35,075.58	-1,098.63	-17,175.10	43,123.44
Applied to removed ratio	1.34	0.94	0.86	1.10

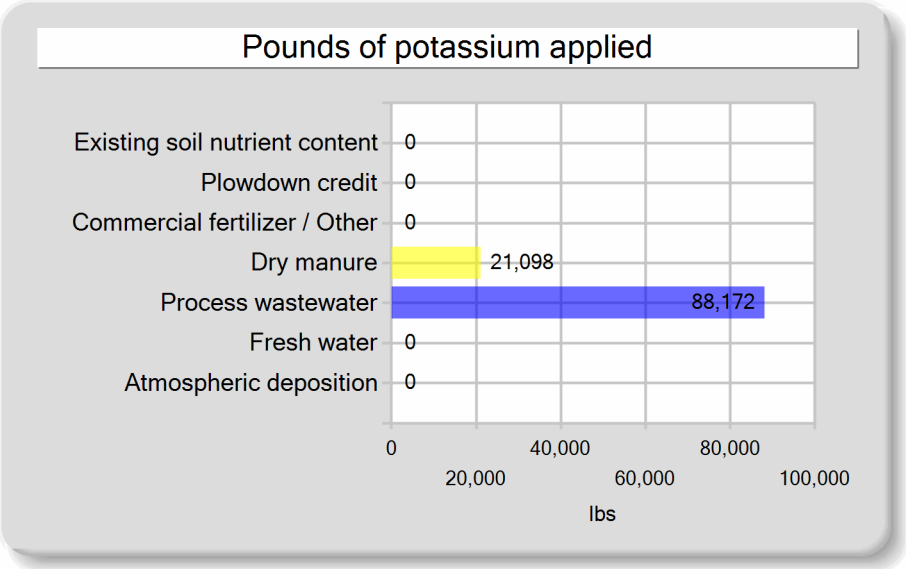
B. POUNDS OF NUTRIENT APPLIED VS. CROP REMOVAL

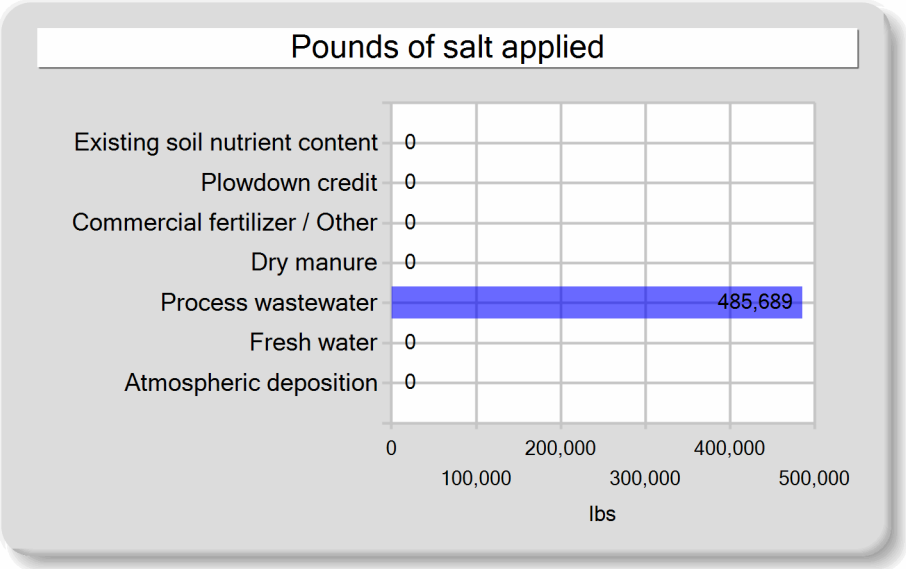


C. POUNDS OF NUTRIENT APPLIED BY MATERIAL TYPE









EXCEPTION REPORTING

A. MANURE, PROCESS WASTEWATER, AND OTHER DAIRY WASTE DISCHARGES

The following is a summary of all manure and process wastewater discharges from the production area to surface water or to land areas (land application areas or otherwise) when not in accordance with the facility's Nutrient Management Plan.

No manure or process wastewater discharges occurred during the reporting period.

B. STORM WATER DISCHARGES

The following is a summary of all storm water discharges from the production area to surface water during the reporting period when not in accordance with the facility 's Nutrient Management Plan.

No stormwater discharges occurred during the reporting period.

C. LAND APPLICATION AREA TO SURFACE WATER DISCHARGES

The following is a summary of all discharges from the land application area to surface water that have occurred during the reporting period when not in accordance with the facility's Nutrient Management Plan.

No land application area to surface water discharges occurred during the reporting period.

NUTRIENT MANAGEMENT PLAN AND EXPORT AGREEMENT STATEMENTS

A. NUTRIENT MANAGEMENT PLAN STATEMENTS

Was the facility's NMP updated in the reporting period?	Yes
Was the facility's NMP developed by a certified nutrient management planner (specialist) as specified in Attachment C of the General Order?	Yes
Was the facility's NMP approved by a certified nutrient management planner (specialist) as specified in Attachment C of the General Order?	Yes

B. EXPORT AGREEMENT STATEMENT

Are there any written agreements with third parties to receive manure or process wastewater that are new or were revised within the reporting period?	No

ADDITIONAL NOTES

A. NOTES

No notes entered for this annual report.

CERTIFICATION

A. OWNER AND/OR OPERATOR CERTIFICATION

I certify under penalty of law that I have personally examined and am familiar with the information submitted in this document and all attachments and that, based on my inquiry of those individuals immediately responsible for obtaining the information, I believe that the information is true, accurate, and complete. I am aware that there are significant penalties for submitting false information, including the possibility of fine and imprisonment.

Jose Soares

SIGNATURE OF OWNER OF FACILITY	SIGNATURE OF OPERATOR OF FACILITY
Jose Soares	SAME AS OWNER
PRINT OR TYPE NAME	PRINT OR TYPE NAME
Jun 25, 2024	
DATE	DATE

ATTACHMENTS

A. REQUIRED ATTACHMENTS

The following lists the required documents that should be attached to the Annual Report when submitted .

Annual Dairy Facility Assessment
Provide an Annual Dairy Facility Assessment (an update to the Preliminary Dairy Facility Assessment in Attachment A) for each reporting period. On the PDFA Final page, click on the ADFA Report button to generate an ADFA report after updating information as needed .

Manure/Process Wastewater Tracking Manifests
Provide copies of all manure/process wastewater tracking manifests for the reporting period, signed by both the owner/operator and the hauler.

Corrective Actions Documents
Provide records documenting any corrective actions taken to correct deficiencies noted as a result of the inspections required in the Monitoring Requirements of the General Order. Deficiencies not corrected in 30 days must be accompanied by an explanation of the factors preventing immediate correction.

Groundwater Monitoring
Dischargers that monitor supply wells or subsurface (tile) drainage systems, or that have monitoring well systems must submit monitoring results as directed in the General Order, Groundwater Reporting Section starting on page MRP-13.

Storm Water Monitoring
Dischargers that are required to monitor storm water more frequently than required in the General Order must submit monitoring results as directed in the General Order, Storm Water Reporting Section on page MRP-14.

Manure / Process Wastewater Tracking Manifest
For
Existing Milk Cow Dairies

General Order No. R5-2007-0035, Attachment D

INSTRUCTIONS

1) Complete one manifest for each hauling event, for each destination. A hauling event may last for several days, as long as the manure is being hauled to the same destination.
2) If there are multiple destinations, complete a separate form for each destination.
3) The operator must obtain the signature of the hauler upon completion of each manure/process wastewater hauling event.
4) The operator shall submit copies of manure/process wastewater tracking manifest(s) with the Annual Monitoring Report for Existing Milk Cow Dairies.

OPERATOR INFORMATION

Name of Operator: Jose Soares

Name of Dairy Facility: Joe & Teresa Soares Dairy #1

Facility Address:

Number and Street	City	County	Zip Code
11076 Vincent RD	Turlock	Merced	95380

Contact Person Name and Phone Number:

Name	Phone Number
Jose Soares	(209) 634-4779

MANURE HAULER INFORMATION

Name of Hauling Company/Person: Cardoso Farm Services Inc

Address of Hauling Company/Person:

Number and Street	City	State	Zip Code
P.O. Box 60	Hilmar	CA	95323

Contact Person:

Name	Phone Number
Arlindo Cardoso	(209) 564-6329

DESTINATION INFORMATION

Composting Facility / Broker / Farmer / Other (identify): Composting Facility

Contact information of Composting Facility, Broker, Farmer, or Other (as identified above):

Name	Phone Number
Cardoso Farm Services Inc	(209) 564-6329

Address	City	State	Zip Code
P.O. Box 60	Hilmar	CA	95323

Destination Address or Assessor's Parcel Number:

Address	City	Zip Code
	Stevinson	95374

Street and nearest cross street (if no address)	County
4th Ave	Merced

Assessor's Parcel Number	Assessor's Parcel Number County

Last date hauled: 05/08/2023

Manure / Process Wastewater Tracking Manifest
For
Existing Milk Cow Dairies

General Order No. R5-2007-0035, Attachment D

MANURE AMOUNT HAULED

Enter the amount of manure hauled in tons, manure solids content, and the method used to calculate the amount:

Manure: 4,095.00 tons

Manure Solids Content: 24.7 %

Method used to determine amount of manure:

TONS PER LOAD

CERTIFICATION

I declare under penalty of law that I personally examined and am familiar with the information submitted in this document, and that based on my inquiry of those individuals immediately responsible for obtaining the information, I believe that the information is true , accurate, and complete. I am aware that there are significant penalties for submitting false information, including the possibility of a fine and imprisonment for knowing violations.

Jose Soares — 5/8/23

Operator Signature — Date

Arlindo Cardoso — 05/08/23

Hauler Signature — Date

Manure / Process Wastewater Tracking Manifest
For
Existing Milk Cow Dairies

General Order No. R5-2007-0035, Attachment D

INSTRUCTIONS

1) Complete one manifest for each hauling event, for each destination. A hauling event may last for several days, as long as the manure is being hauled to the same destination.
2) If there are multiple destinations, complete a separate form for each destination.
3) The operator must obtain the signature of the hauler upon completion of each manure/process wastewater hauling event.
4) The operator shall submit copies of manure/process wastewater tracking manifest(s) with the Annual Monitoring Report for Existing Milk Cow Dairies.

OPERATOR INFORMATION

Name of Operator: Jose Soares

Name of Dairy Facility: Joe & Teresa Soares Dairy #1

Facility Address:

11076 Vincent RD	Turlock	Merced	95380
Number and Street	City	County	Zip Code

Contact Person Name and Phone Number: Jose Soares (209) 634-4779
Name Phone Number

MANURE HAULER INFORMATION

Name of Hauling Company/Person: Cardoso Farm Services Inc

Address of Hauling Company/Person:

P.O. Box 60	Hilmar	CA	95323
Number and Street	City	State	Zip Code

Contact Person: Arlindo Cardoso (209) 564-6329
Name Phone Number

DESTINATION INFORMATION

Composting Facility / Broker / Farmer / Other (identify): Composting Facility

Contact information of Composting Facility, Broker, Farmer, or Other (as identified above):

Cardoso Farm Services Inc (209) 564-6329
Name Phone Number

P.O. Box 60	Hilmar	CA	95323
Address	City	State	Zip Code

Destination Address or Assessor's Parcel Number:

	Stevinson	95374
Address	City	Zip Code

4th Ave	Merced
Street and nearest cross street (if no address)	County

Assessor's Parcel Number Assessor's Parcel Number County

Last date hauled: 11/14/2023

Manure / Process Wastewater Tracking Manifest
For
Existing Milk Cow Dairies

General Order No. R5-2007-0035, Attachment D

MANURE AMOUNT HAULED

Enter the amount of manure hauled in tons, manure solids content, and the method used to calculate the amount:

Manure: 4,160.00 tons

Manure Solids Content: 27.1 %

Method used to determine amount of manure:

tons per load

CERTIFICATION

I declare under penalty of law that I personally examined and am familiar with the information submitted in this document, and that based on my inquiry of those individuals immediately responsible for obtaining the information, I believe that the information is true , accurate, and complete. I am aware that there are significant penalties for submitting false information, including the possibility of a fine and imprisonment for knowing violations.

Jose Soares 11/14/23

Operator Signature Date

Arlindo Cardoso 11/14/23

Hauler Signature Date

DELLAVALLE LABORATORY INC

Report of Dairy Well Water Analysis

Soares & Sons Dairy #1
PO Box 189
Delhi CA 95315
00-0022847 08

E-mail: jtsoaresdairy@aol.com
Copy To: cas.labs@yahoo.com

Lab No.: 23L0453
Sampled By: M. Dekeyrel
Requested By: Manuela Avila
Submitted Date: 12/07/23
Reported Date: 12/13/23
Project:
Crop ID:

		Date Sampled	Time Sampled	EC µmhos/cm	EC mmhos/cm	NO_3-N mg/L	Field NH_4-N mg/L	Total NH_4-N mg/L	pH at 25°C unit
1	DW - Office	12/06/23	12:10	728	0.73	26.6	ND		7.7
2	DW - MB	12/06/23	12:15	794	0.79	31.1	ND		7.8

***** = Field NH_4-N not Taken.
ND = None Detected

Approved By: Scott M Friedland
Laboratory Director\Technical Manager
ELAP Certification #1595
A2LA Certification #6440.02

Report of Dairy Well Water Analysis

DELLAVALLE LABORATORY INC

Soares & Sons Dairy #1
PO Box 189
Delhi CA 95315
00-0022847 08

E-mail: jtsoaresdairy@aol.com
Copy To: cas.labs@yahoo.com

Lab No.: 23L0453
Sampled By: M. Dekeyrel
Requested By: Manuela Avila
Submitted Date: 12/07/23
Reported Date: 12/13/23
Project:
Crop ID:

		Date Sampled	Time Sampled	EC µmhos/cm	EC mmhos/cm	NO_3-N mg/L	Nitrogen agricultural use Calcs calculated from nitrate-N lbs/AcFoot	lbs/AcInch
1	DW - Office	12/06/23	12:10	728	0.73	**26.6**	72.62	6.05
2	DW - MB	12/06/23	12:15	794	0.79	**31.1**	84.90	7.08

Water for Crop Use **Nutrient Management**	Total Salts	Total Salts	Nitrate Nitrogen
Low Levels	< 900	< 0.90	< 2.0
Significant Levels	900-2200	0.90-2.2	2.0-10.0
High Levels	2200+	2.2+	10.0+

Sampling abbreviations: H.B. = hose bib, S.P. = stand pipe, Dom. = domestic well, IR = irrigation well, AG = ag well.
Nitrate exceeds Water Quality MCL levels if results are above 10 mg/l nitrate-nitrogen NO3-N (equivalent to 45 mg/l nitrate, NO3).
Total Salt results are used to monitor changes of salt in the well aquifers between annual tests.
MCL = Maximum Contaminant Level according to the California Domestic Water Quality and Monitoring Regulations (Title 22)
MDL = Method Detection Limit; **RL** = Reporting Limit, mg/L = ppm.
SM = Standard Methods for the Examination of Water and Wastewater
EPA = Environmental Protection Agency methods used unless otherwise indicated.

DELLAVALLE™
LABORATORY INC

Soares & Sons Dairy #1
PO Box 189
Delhi, CA 95315

Account# 00-0022847
Account Manager: Ben Nydam
Submitted By: Manuela Avila

Received: 12/07/2023 15:35
Reported: 12/13/2023 15:14

Samples in this Report

Lab ID	Sample	Matrix	Sampled By	Crop	Date Sampled
23L0453-01	DW - Office	Ag Water	M. Dekeyrel		12/06/2023 12:10
23L0453-02	DW - MB	Ag Water	M. Dekeyrel		12/06/2023 12:15

Default Cooler Temperature on Receipt °C: -3.0
Containers Intact
COC/Labels Agree
Received On Ice

Notes and Definitions

Item	Definition
H	Hold Time Exceeded
MCL	Drinking Water Maximum Contaminant Level
ND	Analyte NOT DETECTED at or above the reporting limit.
NES	Not Enough Sample
*	Not Taken
RPD	Relative Percent Difference
%REC	Percent Recovery
Source	Sample that was matrix spiked or duplicated.

Scott McFriedland

Laboratory Director/Technical Manager

ELAP Certification #1595
A2LA Certification #6440.02

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DELLAVALLE™
LABORATORY INC

Soares & Sons Dairy #1
PO Box 189
Delhi, CA 95315

Account# 00-0022847
Account Manager: Ben Nydam
Submitted By: Manuela Avila

Received: 12/07/2023 15:35
Reported: 12/13/2023 15:14

Sample Results

Sample: DW - Office
23L0453-01 (Water)

Sampled: 12/6/2023 12:10
Sampled By: M. Dekeyrel

Analyte	Result	Units	Reporting Limit	DIL	DW MCL	Date/Time Analyzed	Method	Notes	Batch
Electrical Conductivity	**0.73**	mmhos/cm	0.01	1		12/11/23 16:28	SM 2510 B		BEL0301
Electrical Conductivity umhos	**728**	umhos/cm	10.0	1		12/11/23 16:28	SM 2510 B		BEL0301
Ammonia (as N)	ND	mg/L	0.00	1		12/06/23 12:10	Field		BEL0482
Nitrate Nitrogen as NO3N	**26.6**	mg/L	0.1	1	10	12/08/23 06:54	EPA 300.0		BEL0227
Temperature	**25.0**	units	0.0	1		12/11/23 16:28	SM 4500-H+	H	BEL0301
pH	**7.7**	units	1.0	1		12/11/23 16:28	SM 4500-H+	H	BEL0301

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LABORATORY INC

Soares & Sons Dairy #1
PO Box 189
Delhi, CA 95315

Account# 00-0022847
Account Manager: Ben Nydam
Submitted By: Manuela Avila

Received: 12/07/2023 15:35
Reported: 12/13/2023 15:14

Sample Results
(Continued)

Sample: DW - MB
23L0453-02 (Water)

Sampled: 12/6/2023 12:15
Sampled By: M. Dekeyrel

Analyte	Result	Units	Reporting Limit	DIL	DW MCL	Date/Time Analyzed	Method	Notes	Batch
Electrical Conductivity	**0.79**	mmhos/cm	0.01	1		12/11/23 16:29	SM 2510 B		BEL0301
Electrical Conductivity umhos	**794**	umhos/cm	10.0	1		12/11/23 16:29	SM 2510 B		BEL0301
Ammonia (as N)	ND	mg/L	0.00	1		12/06/23 12:15	Field		BEL0482
Nitrate Nitrogen as NO3N	**31.1**	mg/L	0.1	1	10	12/08/23 07:14	EPA 300.0		BEL0227
Temperature	**25.0**	units	0.0	1		12/11/23 16:29	SM 4500-H+	H	BEL0301
pH	**7.8**	units	1.0	1		12/11/23 16:29	SM 4500-H+	H	BEL0301

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DELLAVALLE™ LABORATORY INC

Soares & Sons Dairy #1
PO Box 189
Delhi, CA 95315

Account# 00-0022847
Account Manager: Ben Nydam
Submitted By: Manuela Avila

Received: 12/07/2023 15:35
Reported: 12/13/2023 15:14

Quality Control

Analyte	Result	Qual	Reporting Limit	Units	Spike Level	Source Result	%REC	%REC Limits	RPD	RPD Limit
Batch: BEL0227										
Blank (BEL0227-BLK1)					Prepared & Analyzed: 12/7/2023					
Nitrate Nitrogen as NO3N	ND		0.1	mg/L						
Blank (BEL0227-BLK2)					Prepared & Analyzed: 12/7/2023					
Nitrate Nitrogen as NO3N	ND		0.1	mg/L						
Blank (BEL0227-BLK3)					Prepared & Analyzed: 12/8/2023					
Nitrate Nitrogen as NO3N	ND		0.1	mg/L						
Blank (BEL0227-BLK4)					Prepared & Analyzed: 12/8/2023					
Nitrate Nitrogen as NO3N	ND		0.1	mg/L						
LCS (BEL0227-BS1)					Prepared & Analyzed: 12/7/2023					
Nitrate Nitrogen as NO3N	5.3		0.1	mg/L	5.000		106	90-110		
LCS (BEL0227-BS2)					Prepared & Analyzed: 12/8/2023					
Nitrate Nitrogen as NO3N	5.1		0.1	mg/L	5.000		102	90-110		
LCS (BEL0227-BS3)					Prepared & Analyzed: 12/8/2023					
Nitrate Nitrogen as NO3N	5.1		0.1	mg/L	5.000		101	90-110		
Duplicate (BEL0227-DUP1)		**Source: 23L0251-01**			Prepared & Analyzed: 12/7/2023					
Nitrate Nitrogen as NO3N	2.6		0.1	mg/L		2.6			0.610	10
Duplicate (BEL0227-DUP2)		**Source: 23L0351-02**			Prepared & Analyzed: 12/8/2023					
Nitrate Nitrogen as NO3N	0.04		0.1	mg/L		0.04			0.00	10
Duplicate (BEL0227-DUP3)		**Source: 23L0446-01**			Prepared & Analyzed: 12/8/2023					
Nitrate Nitrogen as NO3N	3.9		0.1	mg/L		3.9			0.00	10
Matrix Spike (BEL0227-MS1)		**Source: 23L0251-01**			Prepared & Analyzed: 12/7/2023					
Nitrate Nitrogen as NO3N	7.8		0.1	mg/L	5.000	2.6	103	90-110		
Matrix Spike (BEL0227-MS2)		**Source: 23L0351-02**			Prepared & Analyzed: 12/8/2023					
Nitrate Nitrogen as NO3N	5.3		0.1	mg/L	5.000	0.04	105	90-110		
Matrix Spike (BEL0227-MS3)		**Source: 23L0446-01**			Prepared & Analyzed: 12/8/2023					
Nitrate Nitrogen as NO3N	9.1		0.1	mg/L	5.000	3.9	104	90-110		
Reference (BEL0227-SRM1)					Prepared & Analyzed: 12/7/2023					
Nitrate Nitrogen as NO3N	10.1			mg/L	10.00		101	90-110		
Reference (BEL0227-SRM2)					Prepared & Analyzed: 12/7/2023					
Nitrate Nitrogen as NO3N	10.2			mg/L	10.00		102	90-110		

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DELLAVALLE™
LABORATORY INC

Soares & Sons Dairy #1
PO Box 189
Delhi, CA 95315

Account# 00-0022847
Account Manager: Ben Nydam
Submitted By: Manuela Avila

Received: 12/07/2023 15:35
Reported: 12/13/2023 15:14

Quality Control
(Continued)

Analyte	Result	Qual	Reporting Limit	Units	Spike Level	Source Result	%REC	%REC Limits	RPD	RPD Limit
Batch: BEL0227 (Continued)										
Reference (BEL0227-SRM3)					Prepared & Analyzed: 12/8/2023					
Nitrate Nitrogen as NO3N	10.3			mg/L	10.00		103	90-110		
Reference (BEL0227-SRM4)					Prepared & Analyzed: 12/8/2023					
Nitrate Nitrogen as NO3N	10.3			mg/L	10.00		103	90-110		

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Account Manager: Ben Nydam
Submitted By: Manuela Avila

Received: 12/07/2023 15:35
Reported: 12/13/2023 15:14

Quality Control
(Continued)

Analyte	Result	Qual	Reporting Limit	Units	Spike Level	Source Result	%REC	%REC Limits	RPD	RPD Limit
Batch: BEL0301										
Blank (BEL0301-BLK1)					Prepared & Analyzed: 12/11/2023					
Electrical Conductivity	ND		0.01	mmhos/cm						
Temperature	25.0		0.0	units						
Electrical Conductivity umhos	ND		10.0	umhos/cm						
pH	5.6		1.0	units						
Blank (BEL0301-BLK2)					Prepared & Analyzed: 12/11/2023					
Temperature	25.0		0.0	units						
Electrical Conductivity	ND		0.01	mmhos/cm						
pH	7.6		1.0	units						
Electrical Conductivity umhos	ND		10.0	umhos/cm						
Blank (BEL0301-BLK3)					Prepared & Analyzed: 12/11/2023					
Electrical Conductivity	ND		0.01	mmhos/cm						
Temperature	25.0		0.0	units						
Electrical Conductivity umhos	ND		10.0	umhos/cm						
pH	7.3		1.0	units						
Duplicate (BEL0301-DUP1)		**Source: 23L0451-01**			Prepared & Analyzed: 12/11/2023					
Electrical Conductivity	0.48		0.01	mmhos/cm		0.48			0.0421	10
Electrical Conductivity umhos	476		10.0	umhos/cm		475			0.0421	10
pH	7.9		1.0	units		7.9			0.381	10
Duplicate (BEL0301-DUP2)		**Source: 23L0453-02**			Prepared & Analyzed: 12/11/2023					
Electrical Conductivity	0.80		0.01	mmhos/cm		0.79			0.126	10
Electrical Conductivity umhos	795		10.0	umhos/cm		794			0.126	10
pH	7.8		1.0	units		7.8			0.129	10
Reference (BEL0301-SRM1)					Prepared & Analyzed: 12/11/2023					
Electrical Conductivity	437			umhos/cm	426.0		103	90-110		
Reference (BEL0301-SRM2)					Prepared & Analyzed: 12/11/2023					
pH	7.5			units	7.520		100	67021-101.32		
Reference (BEL0301-SRM3)					Prepared & Analyzed: 12/11/2023					
Electrical Conductivity	1050			umhos/cm	1000		105	90-110		
Electrical Conductivity umhos	1050			umhos/cm	1000		105	90-110		
Reference (BEL0301-SRM4)					Prepared & Analyzed: 12/11/2023					
Electrical Conductivity	1070			umhos/cm	1000		107	90-110		
Electrical Conductivity umhos	1070			umhos/cm	1000		107	90-110		
Reference (BEL0301-SRM5)					Prepared & Analyzed: 12/11/2023					
Electrical Conductivity	1050			umhos/cm	1000		105	90-110		

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LABORATORY INC

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PO Box 189
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Account# 00-0022847
Account Manager: Ben Nydam
Submitted By: Manuela Avila

Received: 12/07/2023 15:35
Reported: 12/13/2023 15:14

Quality Control
(Continued)

Analyte	Result	Qual	Reporting Limit	Units	Spike Level	Source Result	%REC	%REC Limits	RPD	RPD Limit
Batch: BEL0301 (Continued)										
Reference (BEL0301-SRM5)					Prepared & Analyzed: 12/11/2023					
Electrical Conductivity umhos	1050			umhos/cm	1000		105	90-110		
Reference (BEL0301-SRM6)					Prepared & Analyzed: 12/11/2023					
pH	4.0			units	4.000		100	97.5-102.5		
Reference (BEL0301-SRM7)					Prepared & Analyzed: 12/11/2023					
pH	4.0			units	4.000		100	97.5-102.5		
Reference (BEL0301-SRM8)					Prepared & Analyzed: 12/11/2023					
pH	4.0			units	4.000		99.8	97.5-102.5		

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12/07/23 15:35 23L0453

JG

DELLAVALLE LABORATORY, INC.

1910 W. McKinley Avenue, Suite 110 • Fresno, CA 93728
www.dellavallelab.com 559 233-6129 • 800 228-9896 • Fax 559 268-8174

WATER WORK REQUEST

Bill To: Acct No. 22847 Cons. 8

Purchase Order No. _____ Results Needed By _____

Client: Soares Dairy #1, Jose D.
Address: P.O. Box 189
City, State, Zip: Delhi, CA 95315
Phone: _____ Fax: _____
Cell/Email: _____

Copy to: Cardoso Ag Sevices - cas.labs@yahoo.com

Requested by: Manuela Avila

Ranch: _____

Date sampled: 12/6/23

Sampled by: M Dekeuvel

[X] QA/QC Document [X] Copy of Chain [ ] RWQCB

No. of Samples: 2 No. Bottles: 2

Water Type: [ ] Drinking [ ] Wastewater
[X] Ag Water [ ] Ground Water [ ] Mon. Well
[ ] Supply Water [ ] Other _____

Analysis and Bottles Required: *(Please Indicate Analysis)*

(X) DWW1: (EC, pH, NO_3-N, **NH_4-N Field Test)**
(1) l L plastic, unpreserved (white)

( ) DWW2: (DWW1 Plus SO_4, CO_3, HCO_3, Cl, Ca, Mg, Na, TDS)
(1) l L plastic, unpreserved (white)

( ) DCW1: (EC, NO_3-N, TDS)
(1) l L plastic, unpreserved (white)

( ) DPW1: (EC, pH, NO_3-N, NH_4-N, TKN, TDS, TP, TK)
(1) l L plastic, unpreserved (white)

( ) DPW2: (DPW1 Plus Ca, Mg, Na, HCO_3, CO_3, SO_4, Cl)
(1) l L plastic, unpreserved (white)

( ) Other _____

DESCRIPTION OF SAMPLES		Date Sampled	Time Sampled	Field NH4-N (mg/L)	Received Temp °C
1. DW-office	Sampled From:	12/6/23	1210p	Ø	3.0
2. DW-MB	Sampled From:	12/6/23	1215p	Ø	-2.8
3.	Sampled From:				
4.	Sampled From:				
5.	Sampled From:				
6.	Sampled From:				
7.	Sampled From:				
8.	Sampled From:				
9.	Sampled From:				
10.	Sampled From:				

IR Thermometer SN: 221314357
Correction Factor: 0°C
Calibration Due: 03/06/2024
Location: Laboratory

CHAIN OF CUSTODY

Carrier	Signature	Company	Received (Date/Time)	Relinquished (Date/Time)
First	M Dekeuvel	CAS INC	12/6/23 1215p	12/7/23 11A
Second	JLF	DLV	12-7-23 11:00	12-7-23 3:15
Third				
Fourth	[signature]	DLI	12-7-23 15:35	

I guarantee that as the client, or on behalf of the client named, I have the authority to contract the above requested services. Should it be found that I do not have such authority, I agree to be personally liable for all costs and, if there should be action against me for this breach, reasonable attorneys' fees. It is understood that payment is expected to be cash with samples unless terms have been previously arranged. Terms are net 30 days; overdue accounts will be charged a dated damage fee of 2% per month (annually 24 %) or $5.00 per month whichever is greater.

If payment is not made when due and a legitimate dispute exists concerning the product or services of Dellavalle Laboratory, Inc., it will be submitted to mediation under the Rules and Procedures of Creative Alternative to Litigation, Inc. (cal). If the dispute is not resolved in mediation, then the dispute will be submitted to binding arbitration through cal under its Rules and Procedures. The prties will equally bear the costs of mediation/arbitration. If, however, the mediator declares that no legitimate dispute exists, then debtor will pay all mediation and arbitration costs, and in the event of arbitration, reasonable attorneys' fees of Dellavalle Laboratory.

Invoicing Information:

Sampling Hrs ____ Miles ____ Consulting ____

Amt Paid ____ Rec By ____ Check No. ____

Shipping
$ _____ In
$ _____ Out
Date _____

Signature _____

Sample received in cooler with ice?
[ ] Yes [ ] No

ctl:update 2020

Shipping Information: Shipped In ☑ Picked-Up ☐ Walk In ☐ DLI Sampler ☐ Other ☐ _____

Container: Ice Chest ☑ Box ☐ None ☐ | **Refrigerant:** Wet Ice ☐ Blue Ice ☑ None ☐

Samples Preserved with HNO_3 or H_2SO_4 were: ☐ Received Preserved ☐ Preserved Upon Receipt at Laboratory

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Type of Container(s) Received		Sample Number 1	2	3	4	5	6	7	8	9	10
Sample Containers for Internal (DLI) Use *(Containers that go into the Lab)*											
Plastics	100 mL sterile plastic $Na_2S_2O_3$ (Green)										
	250 mL unpreserved (White) Plastic										
	250 mL HNO_3 (Red) Plastic										
	* pH Value										
	250 mL H_2SO_4 (Yellow) Plastic										
	* pH Value										
	500 mL unpreserved (White) Plastic										
	1 L unpreserved (White) Plastic										
	1 L unpreserved (BOD) (Purple) Plastic										
"	500mL unpreserved (White) Glass										
	PO4-P Kit										
	Other:										
Sample Containers for Subcontracted ("Send Out") Analyses *(Containers that go in the Subcontract ("Send Out") Refrigerator)*											
Plastics	100 mL sterile plastic $Na_2S_2O_3$ (Green)										
	250 mL unpreserved (White) Plastic										
	250 mL HNO_3 (Red) Plastic										
	250 mL H_2SO_4 (Yellow) Plastic										
	500 mL HNO_3 (Red)										
	1 L unpreserved (White) Plastic										
	1 L unpreserved (BOD) (Purple) Plastic										
	1 L HNO_3 (Red)										
VOA Vials	40 mL AG VOA, $Na_2S_2O_3$ + MCAA										
	40 mL AG VOA, $Na_2S_2O_3$										
	40mL AG VOA unpreserved (White) (Set of 3										
	40 mL AG VOA, $Na_2S_2O_3$ (Green) (Set of 3)										
	40mL VOA, H_3PO_4 (Set of 3)										
	40 mL VOA, HCl (Blue) (Set of 3)										
	40 mL VOA, $Na_2S_2O_3$ (Green) (Set of 3)										
Glass	250 mL AG unpreserved (White)										
	250 mL AG H_2SO_4 (Yellow)										
	250 mL AG $Na_2S_2O_3$ (Green)										
	250 mL AG $Na_2S_2O_3$ + MCAA										
	500 mL glass unpreserved (White)										
	500 mL AG HCl (Blue)										
	1 L AG unpreserved (White)										
	1 L AG H_2SO_4 (Yellow)										
	1 L AG $Na_2S_2O_3$ (Green)										
	1 L AG HCl (Blue)										
Special	Cr^{6+} - 50mL Plastic w/Borate/HCO_3/CO_3										
	Cyanide - 500 mL NaOH										
	Asbestos - 1L P wrapped in foil (Set of 2)										
	Sulfide - 1 L **AG** or **P** NaOH + ZnAc										
	Chlorite/Bromate - 250 mL AG with EDA										
	HAA5 - 250mL AG Ammonium Chlorite										
	DO KIT										
	Other:										
	Other:										

SEND OUT